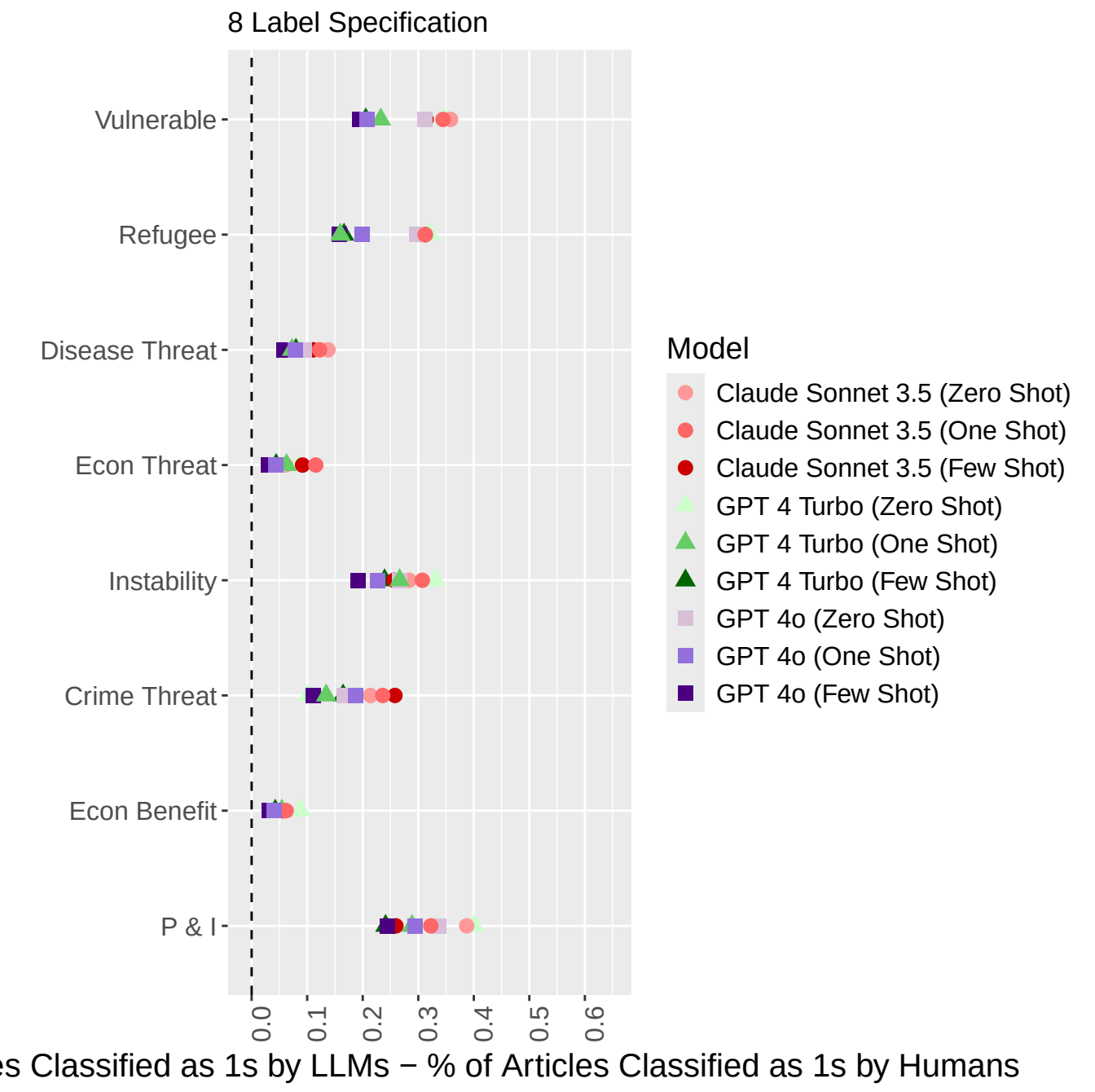
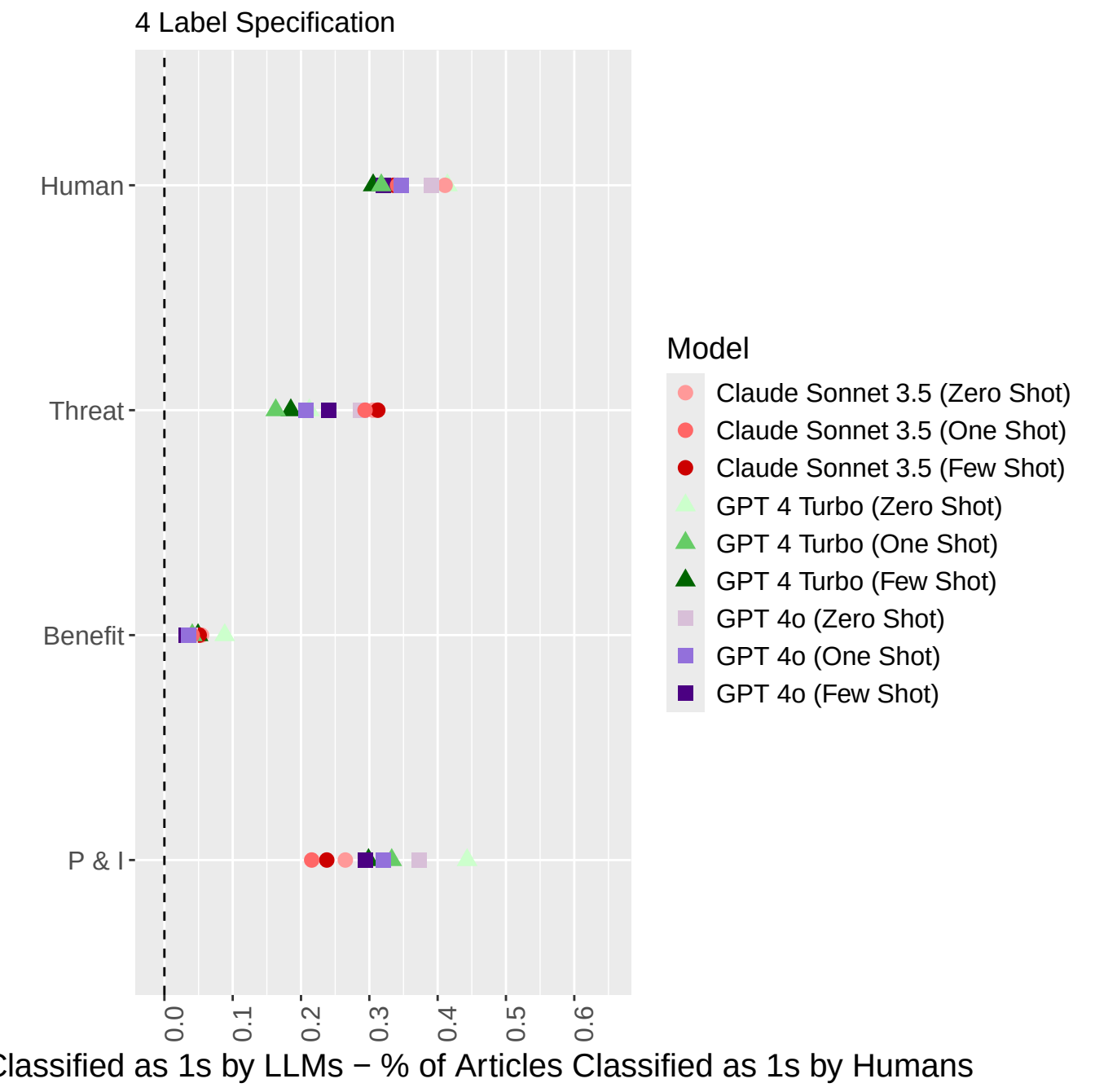
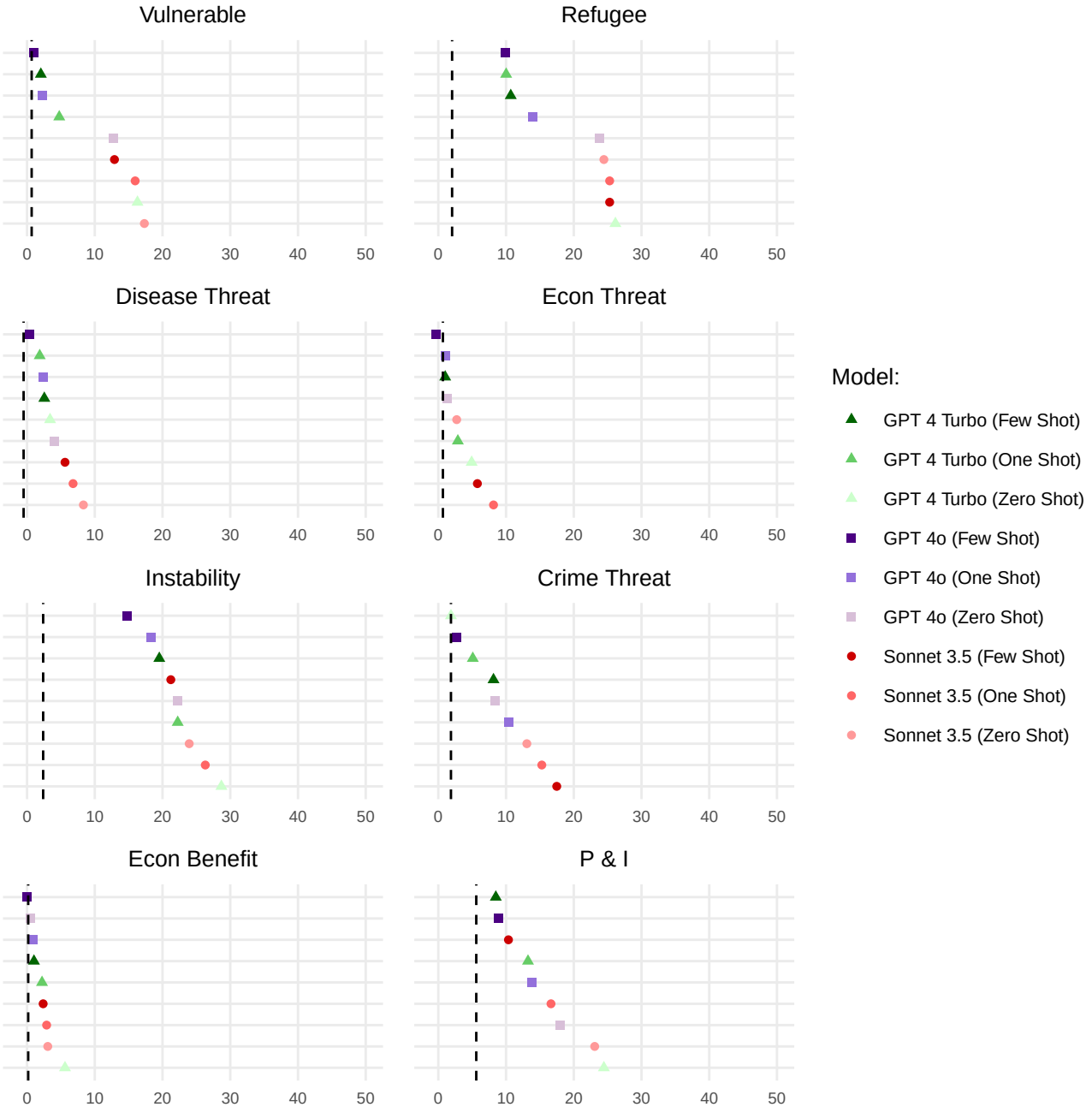
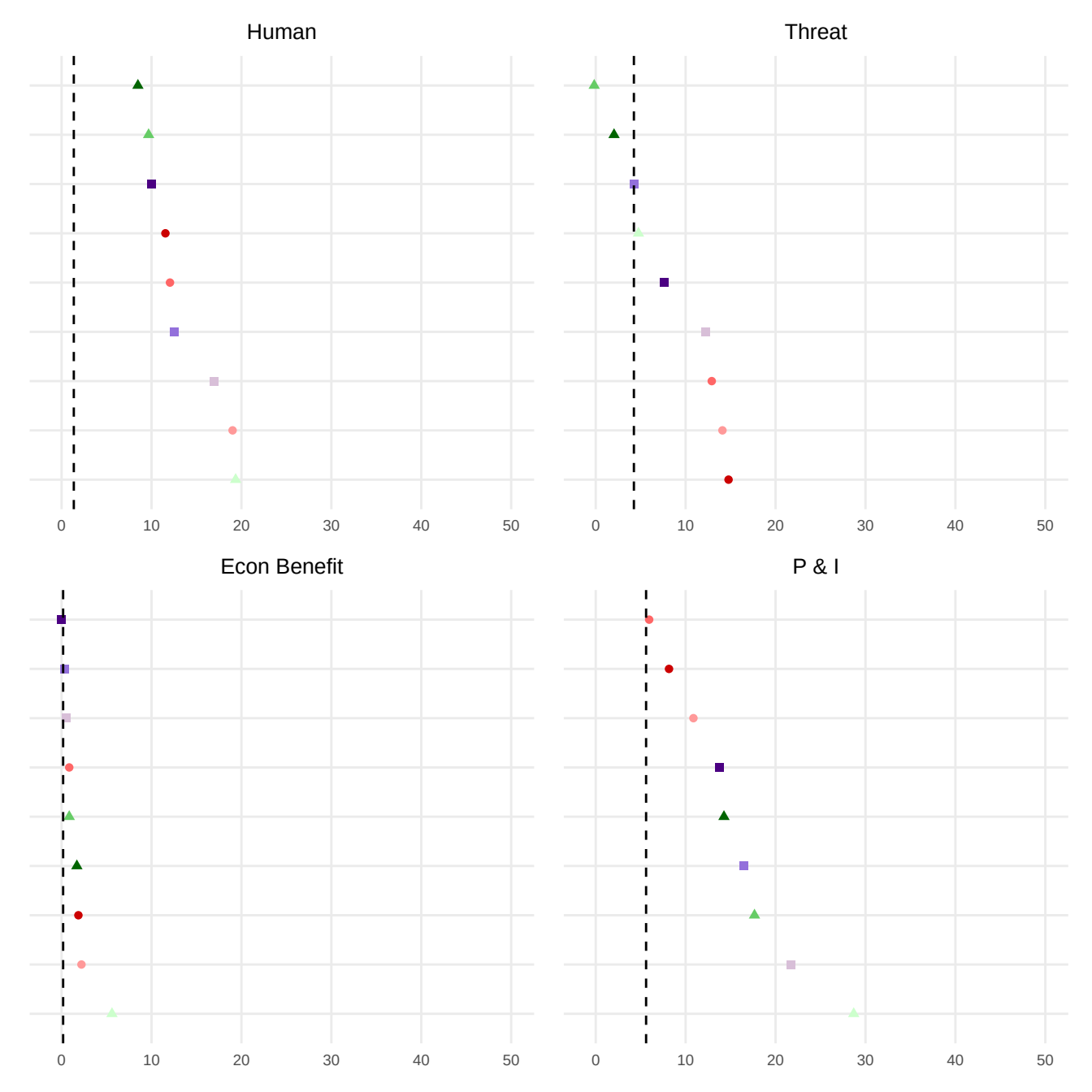

8 Label Specification
Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Model
Claude Sonnet 3.5 (Zero Shot)
Claude Sonnet 3.5 (One Shot)
Claude Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)
es Classified as 1s by LLMs − % of Articles Classified as 1s by Humans

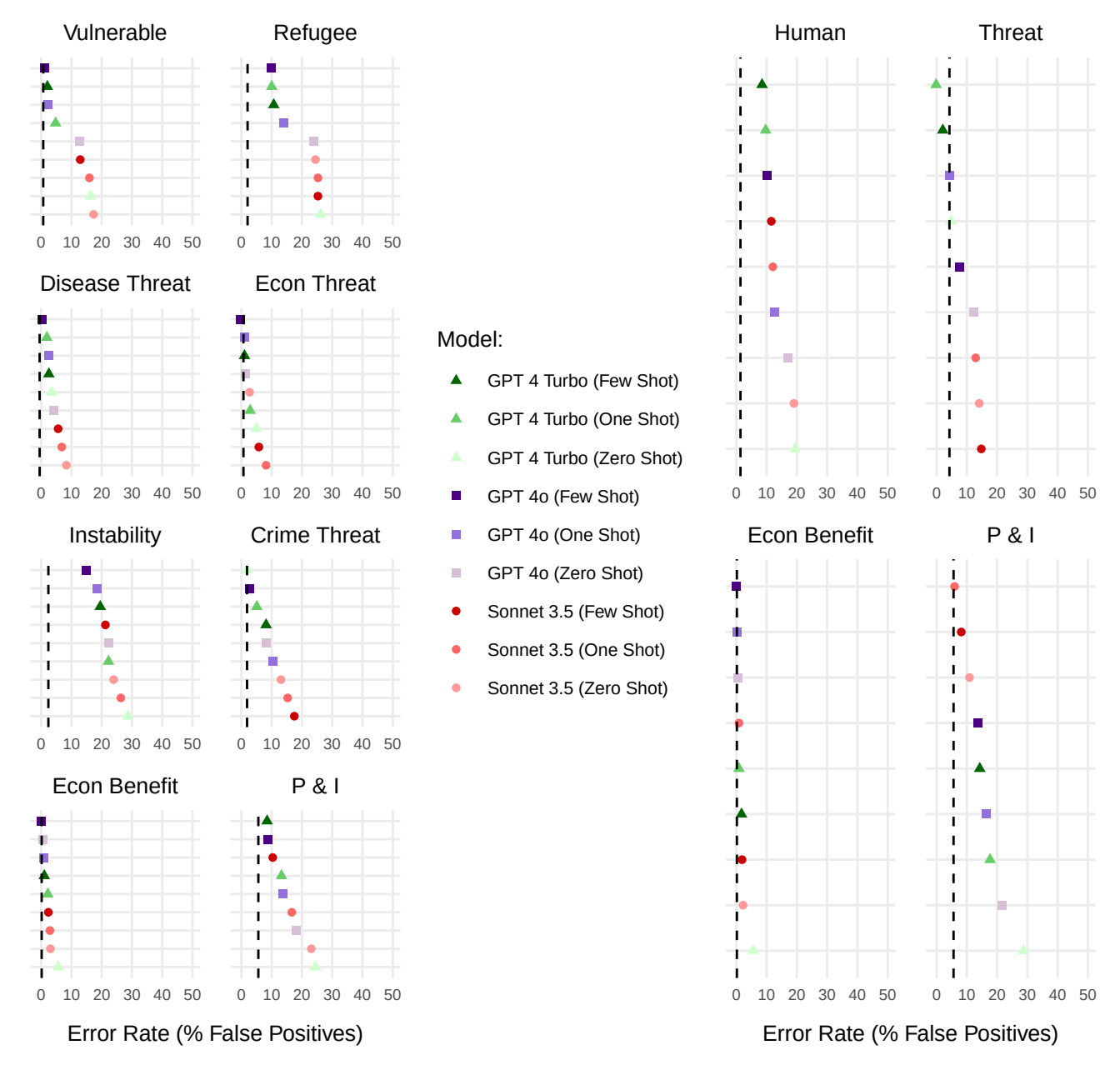

4 Label Specification
Human
Threat
Benefit
P & I
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Model
Claude Sonnet 3.5 (Zero Shot)
Claude Sonnet 3.5 (One Shot)
Claude Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)
Classified as 1s by LLMs − % of Articles Classified as 1s by Humans

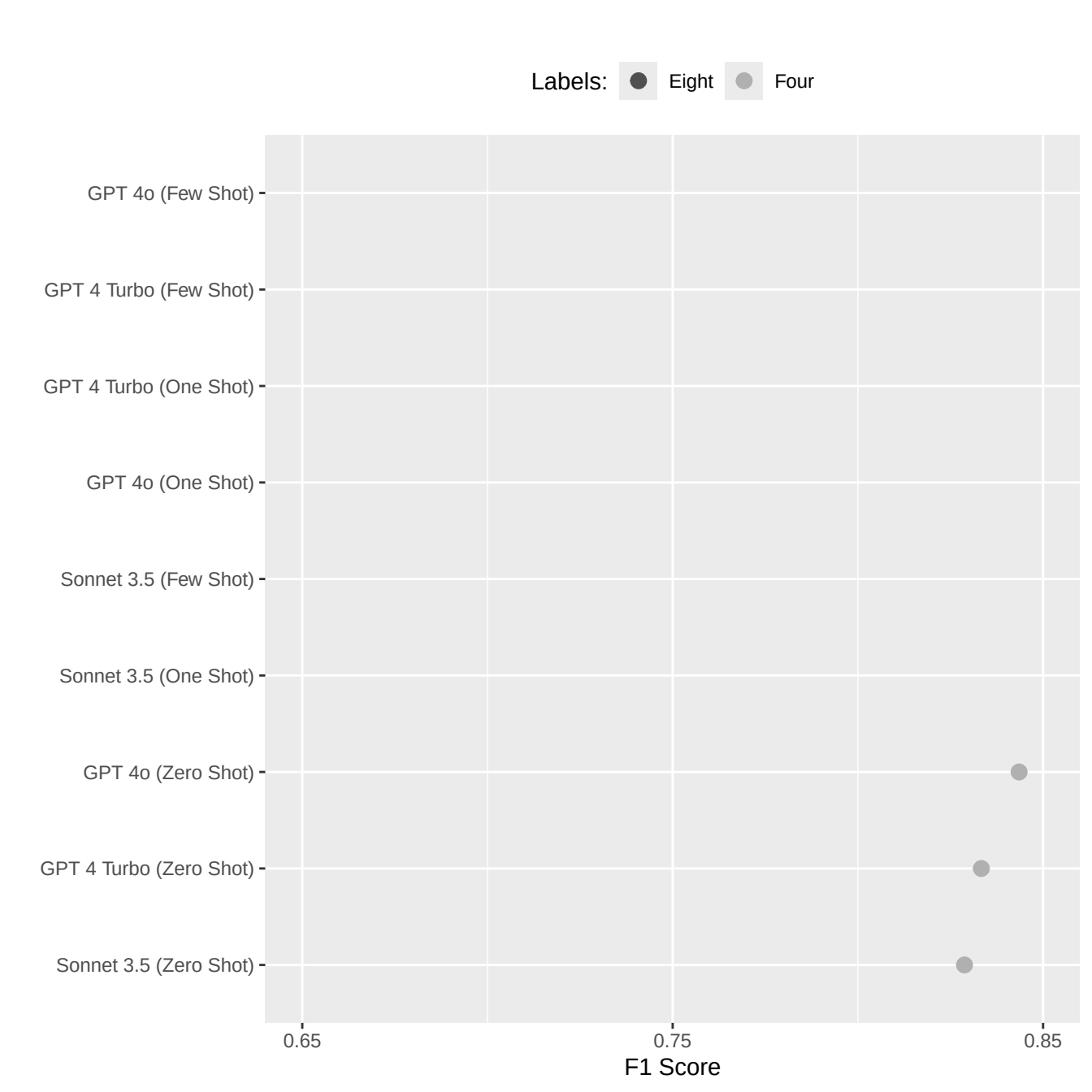


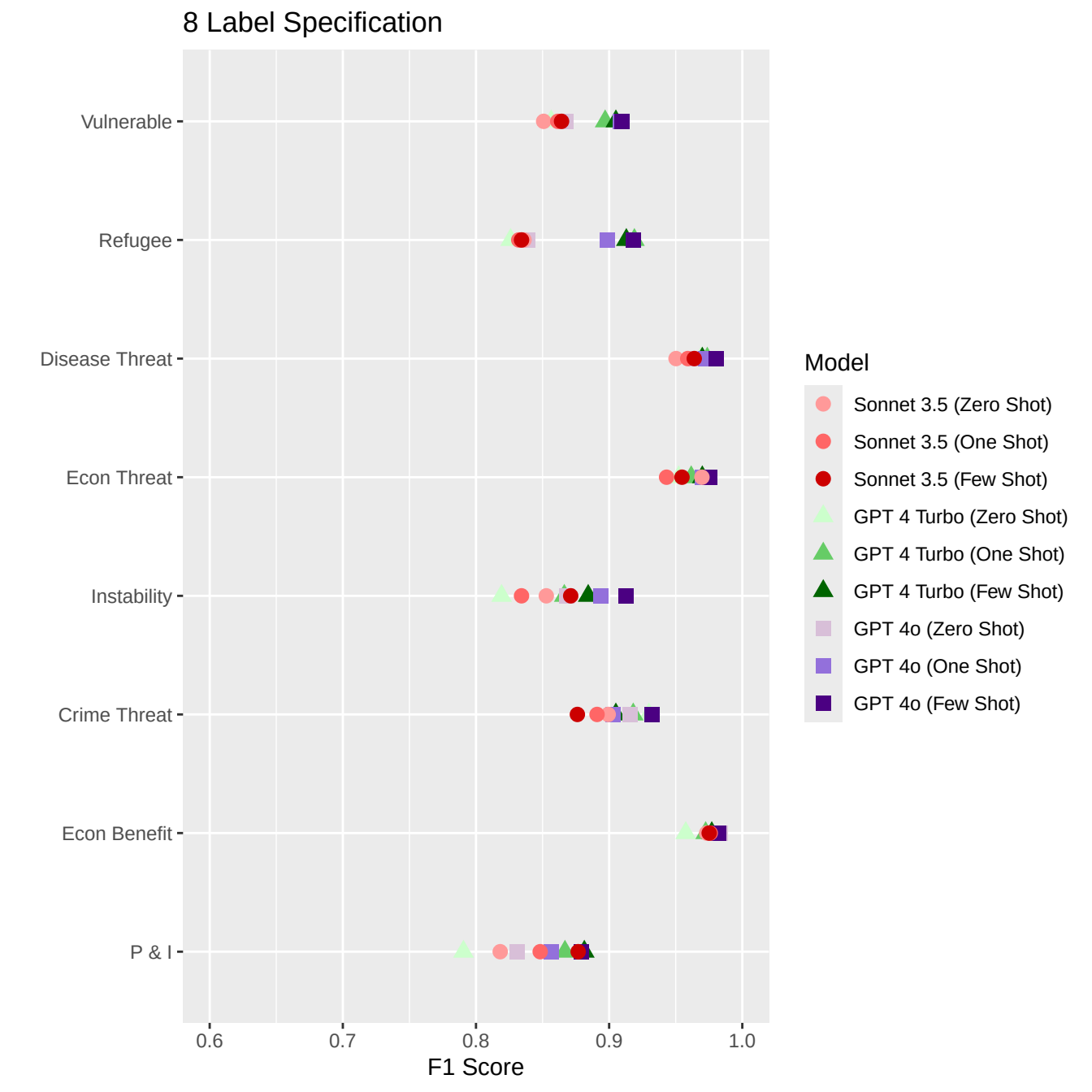
es Classified as 1s by LLMs − % of Articles Classified as 1s by Humans

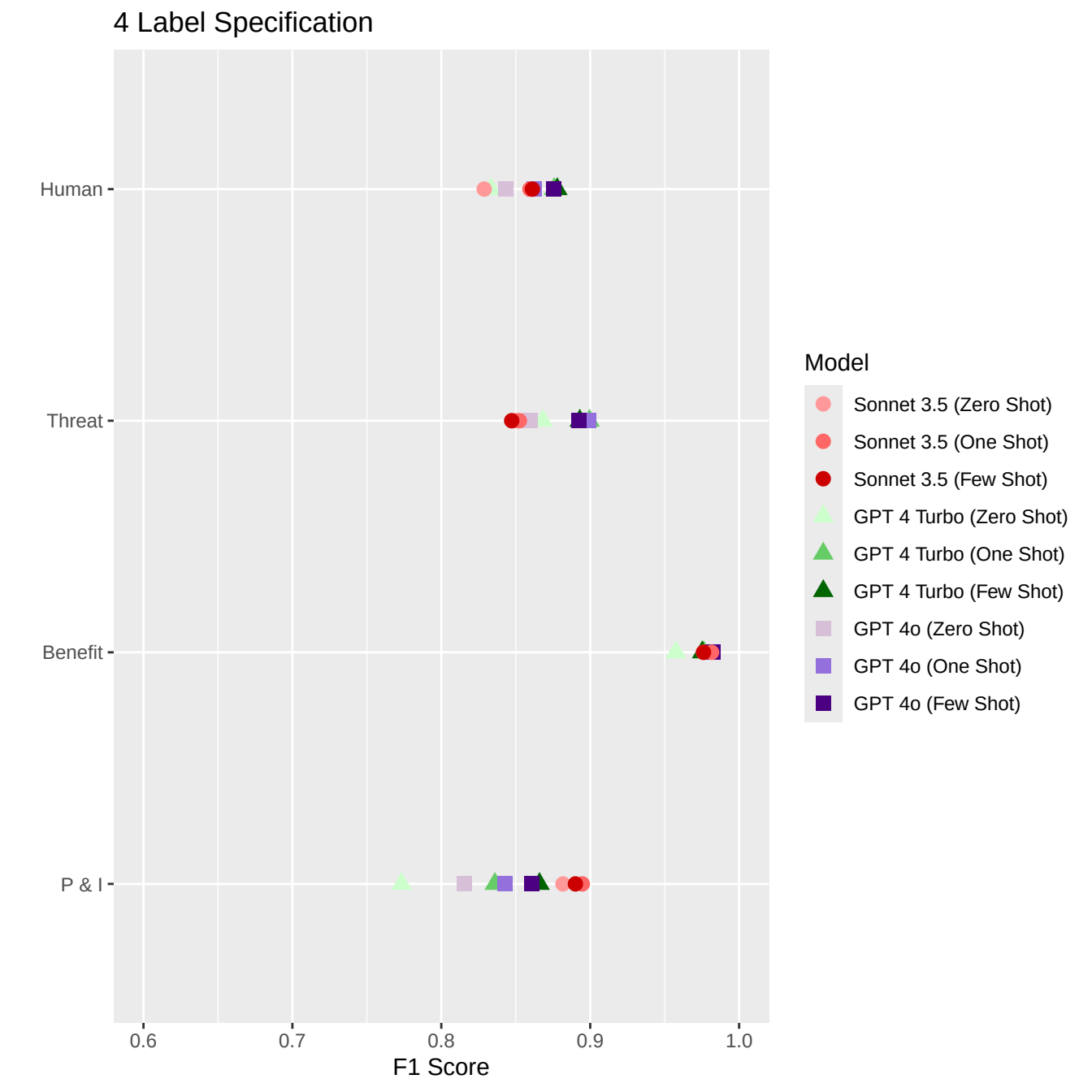

Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
0
10
20
30
40
50
Model:
GPT 4 Turbo (Few Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Zero Shot)
GPT 4o (Few Shot)
GPT 4o (One Shot)
GPT 4o (Zero Shot)
Sonnet 3.5 (Few Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Zero Shot)

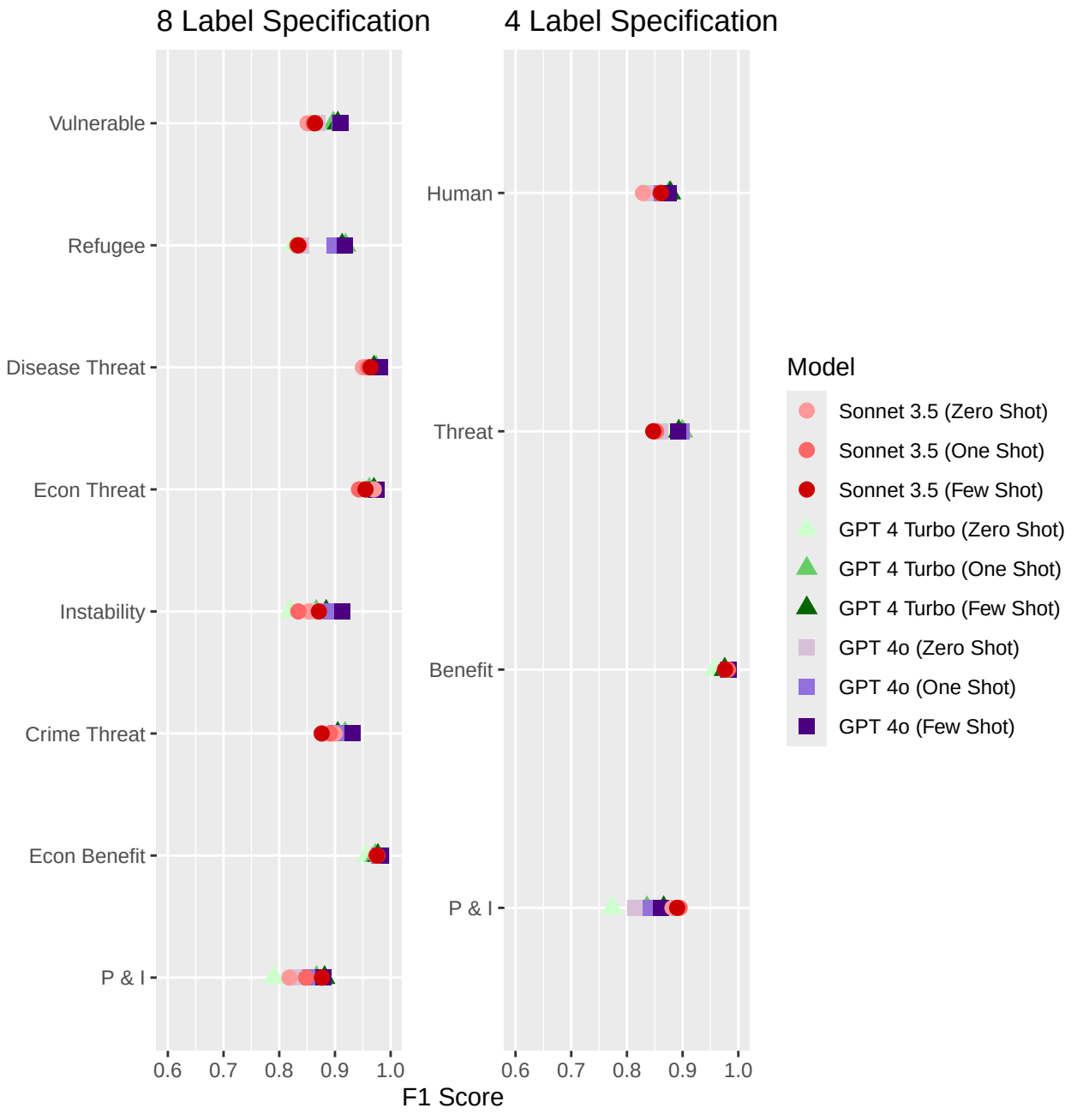

Human
Threat
Econ Benefit
P & I
0
10
20
30
40
50

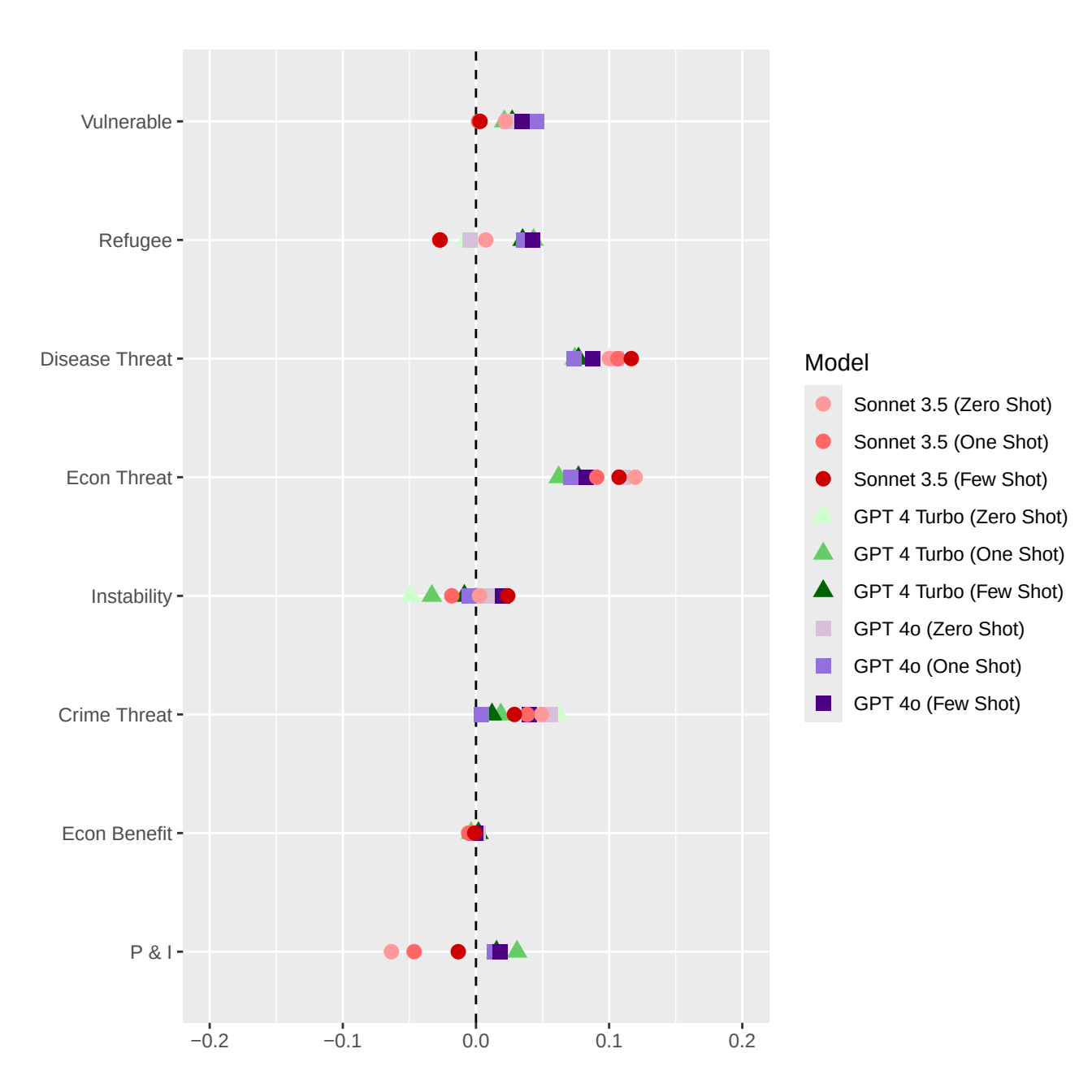

Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
Human
Threat
Econ Benefit
P & I
0 10 20 30 40 50
Model:
GPT 4 Turbo (Few Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Zero Shot)
GPT 4o (Few Shot)
GPT 4o (One Shot)
GPT 4o (Zero Shot)
Sonnet 3.5 (Few Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Zero Shot)
Error Rate (% False Positives)
Error Rate (% False Positives)

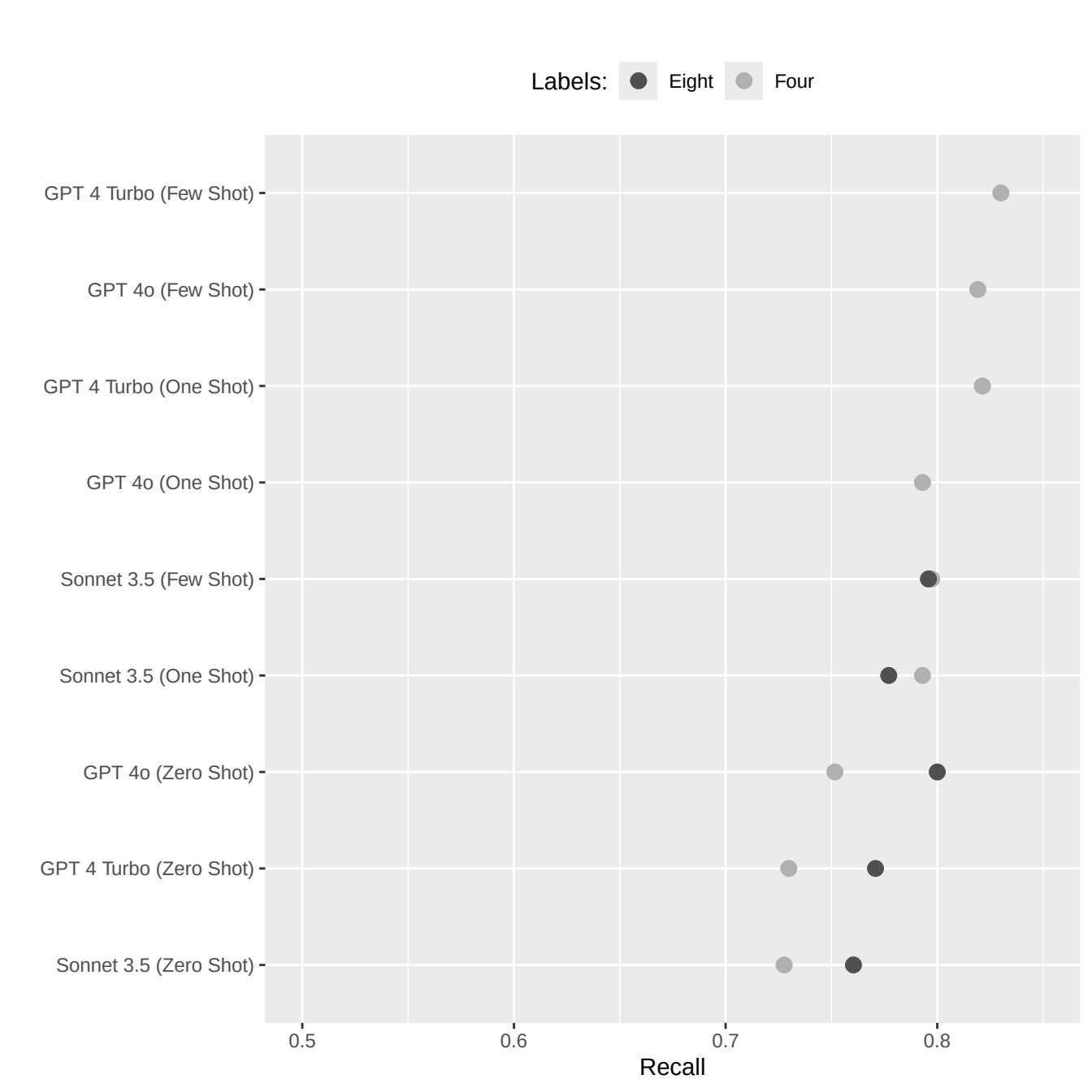

Labels:
Eight
Four
GPT 4o (Few Shot)
GPT 4 Turbo (Few Shot)
GPT 4 Turbo (One Shot)
GPT 4o (One Shot)
Sonnet 3.5 (Few Shot)
Sonnet 3.5 (One Shot)
GPT 4o (Zero Shot)
GPT 4 Turbo (Zero Shot)
Sonnet 3.5 (Zero Shot)
0.65
0.75
0.85
F1 Score

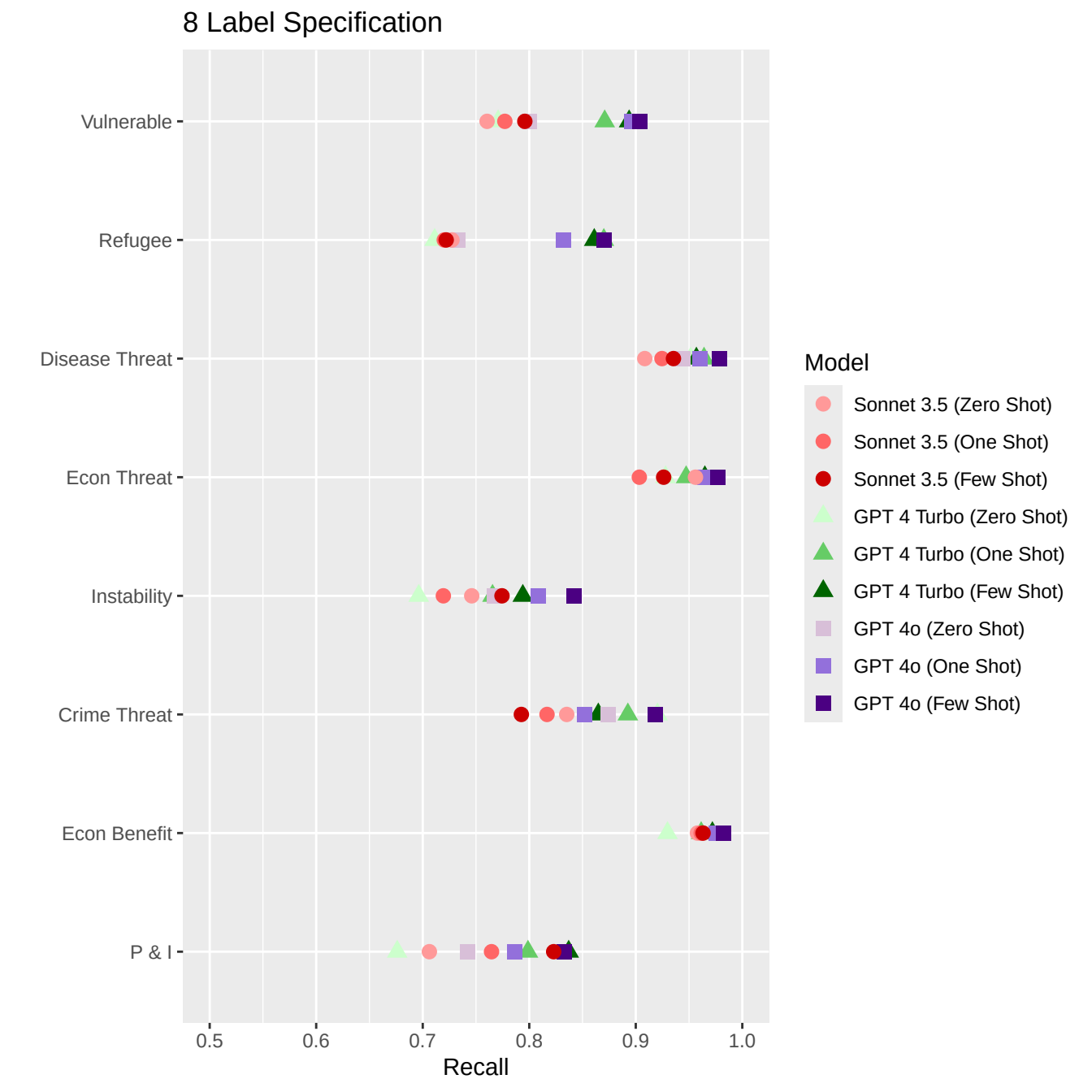

8 Label Specification
Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
0.6
0.7
0.8
0.9
1.0
F1 Score
Model
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)

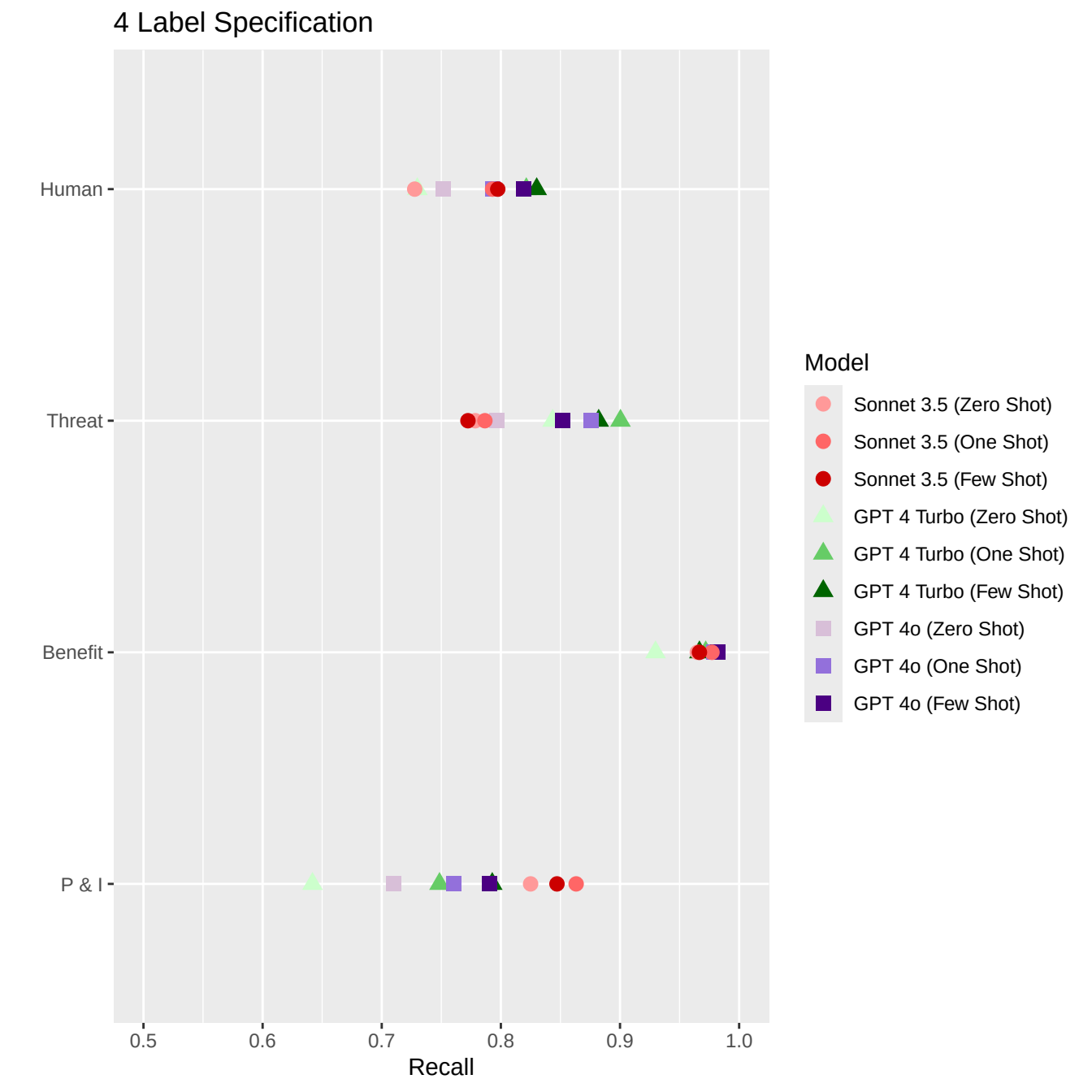

4 Label Specification
Human
Threat
Benefit
P & I
0.6
0.7
0.8
0.9
1.0
F1 Score
Model
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)

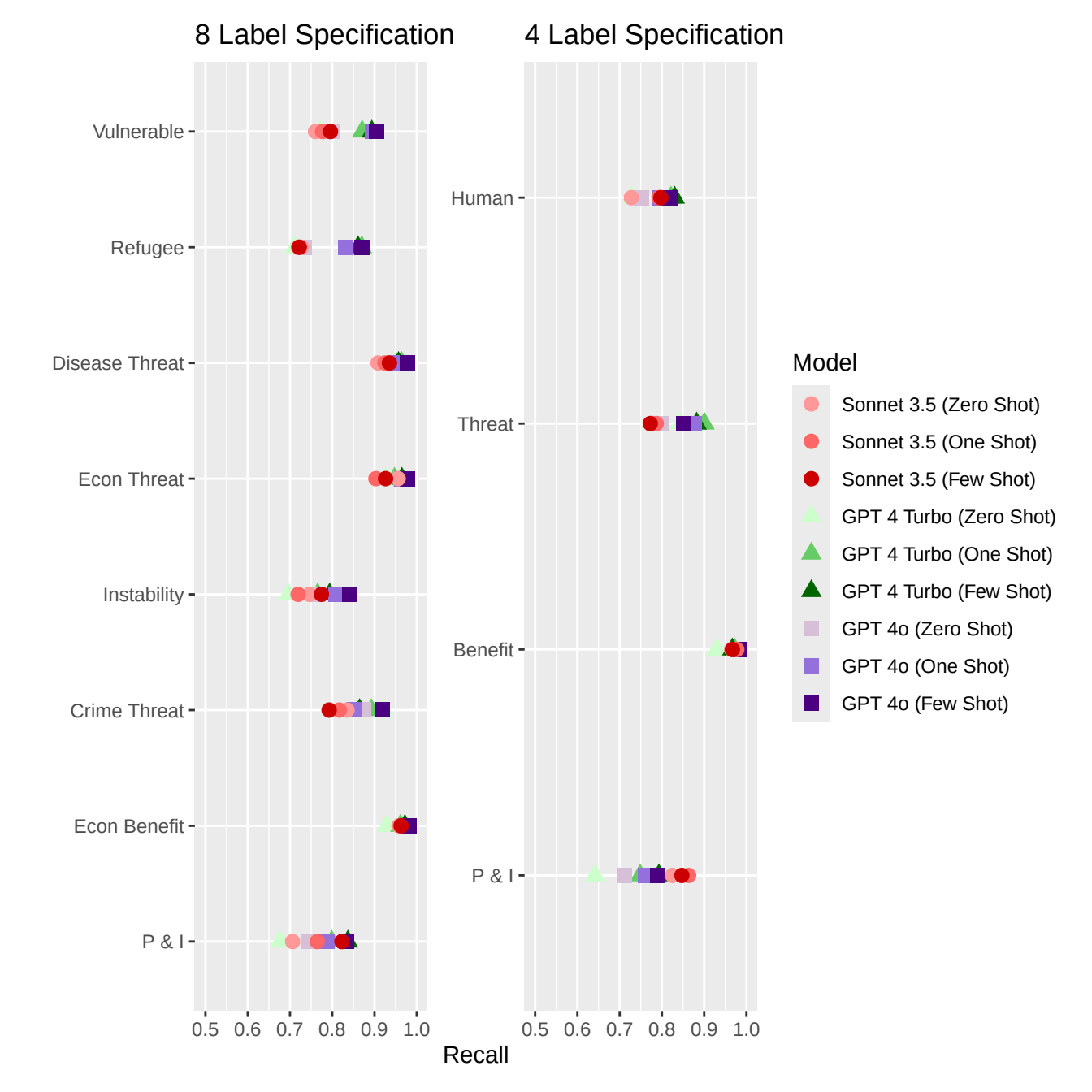

8 Label Specification
4 Label Specification
Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
Human
Threat
Benefit
P & I
0.6
0.7
0.8
0.9
1.0
F1 Score
Model
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)

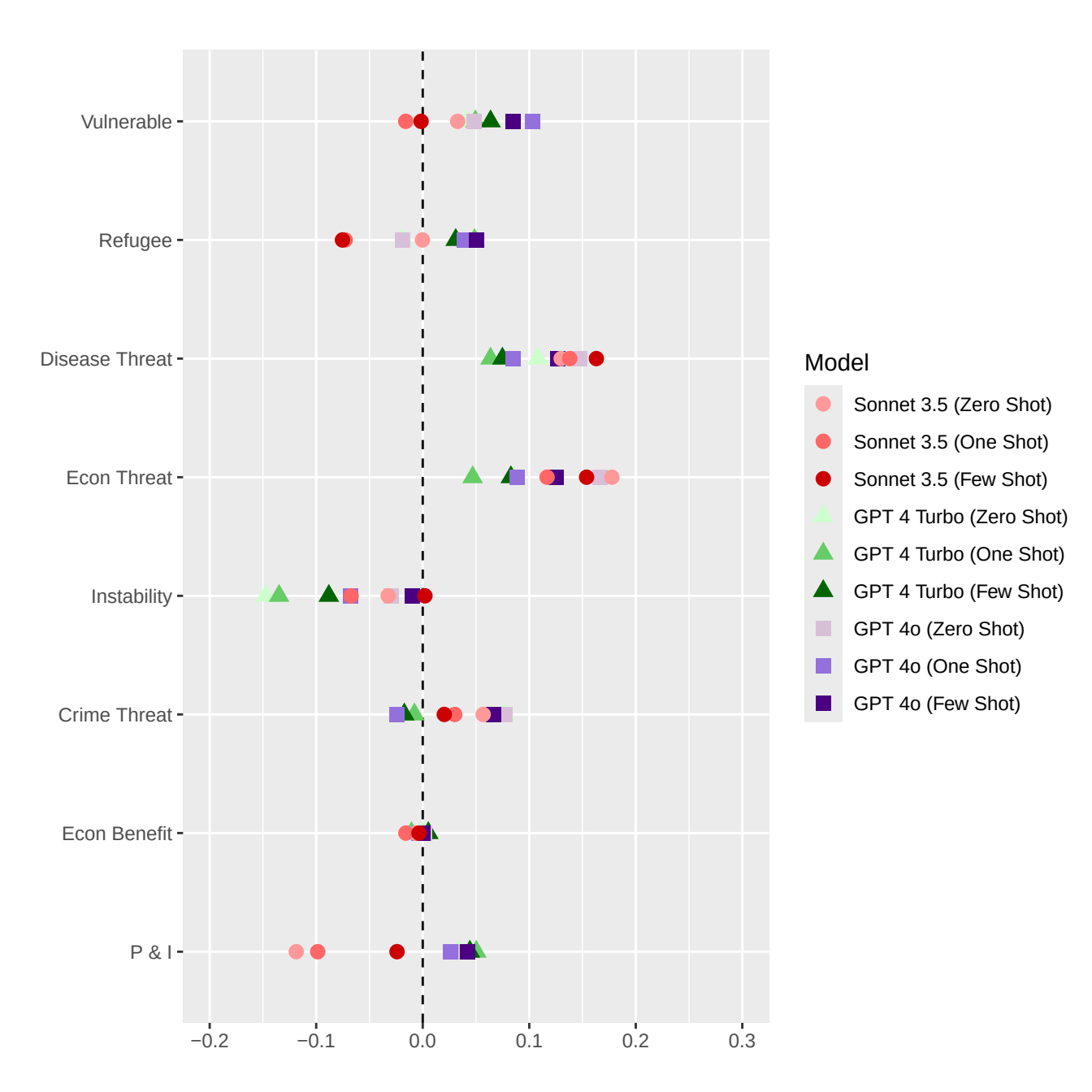

Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
−0.2
−0.1
0.0
0.1
0.2
Model
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)

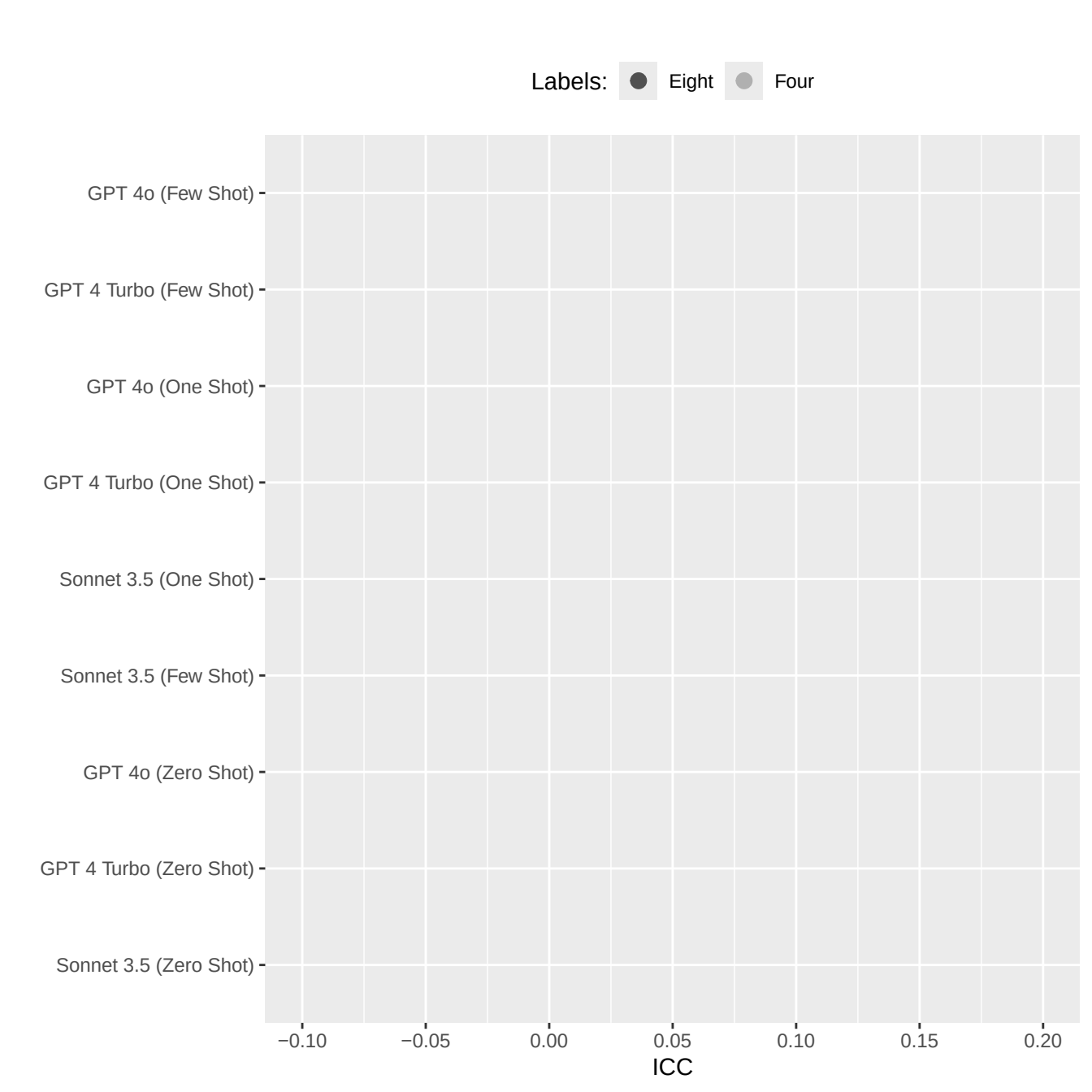

Labels:
Eight
Four
GPT 4 Turbo (Few Shot)
GPT 4o (Few Shot)
GPT 4 Turbo (One Shot)
GPT 4o (One Shot)
Sonnet 3.5 (Few Shot)
Sonnet 3.5 (One Shot)
GPT 4o (Zero Shot)
GPT 4 Turbo (Zero Shot)
Sonnet 3.5 (Zero Shot)
0.5
0.6
0.7
0.8
Recall

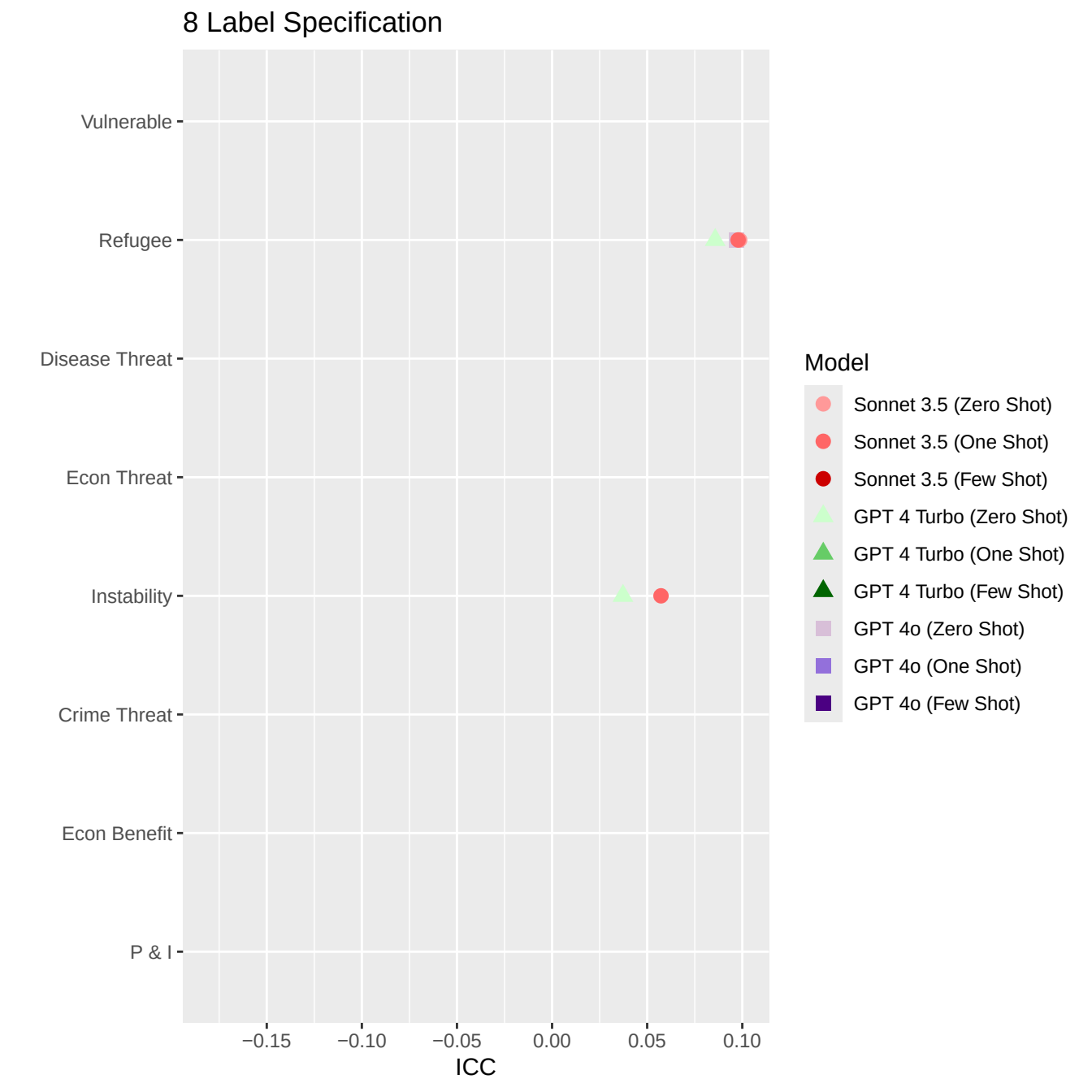

8 Label Specification
Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
0.5
0.6
0.7
0.8
0.9
1.0
Recall
Model
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)

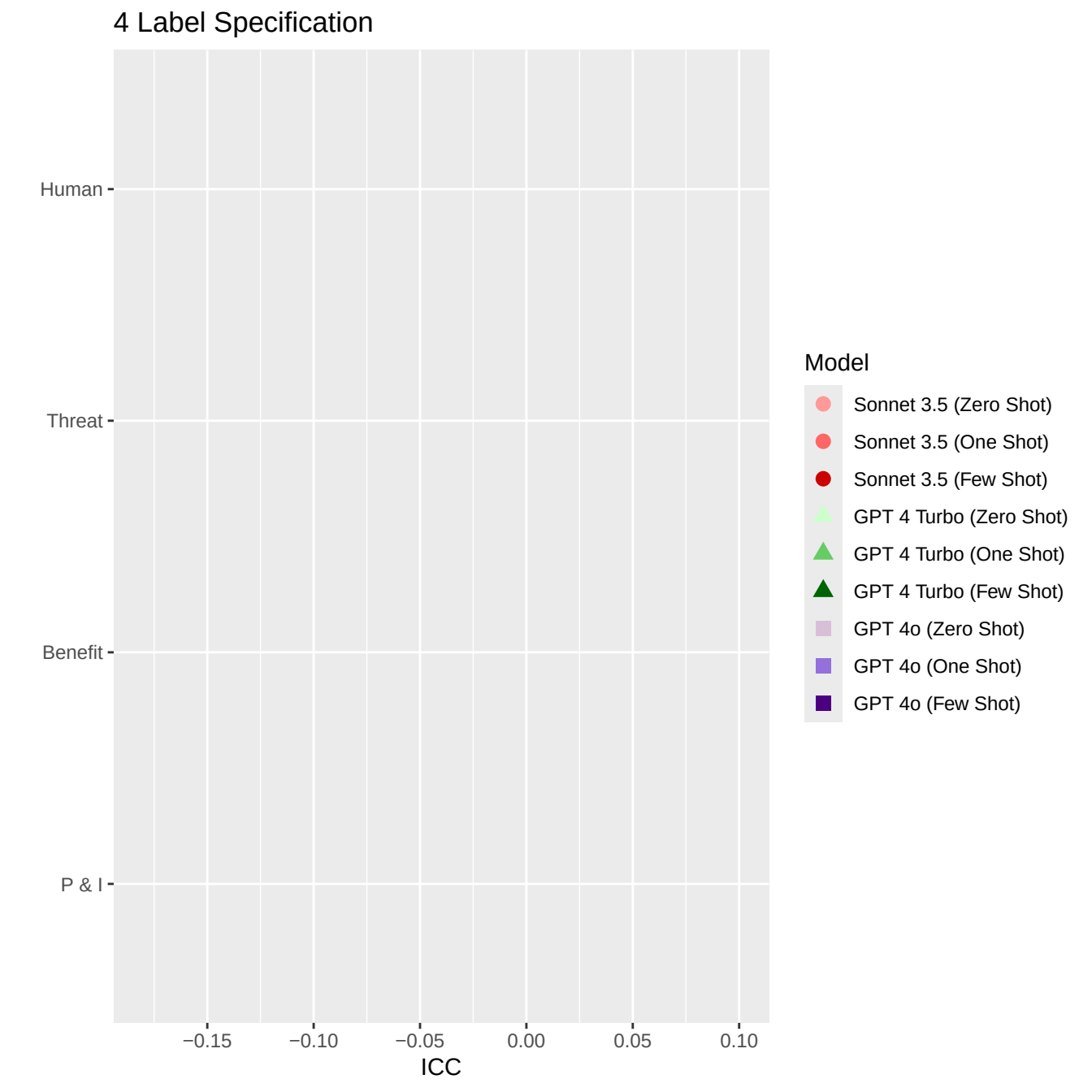

4 Label Specification
Human
Threat
Benefit
P & I
0.5
0.6
0.7
0.8
0.9
1.0
Recall
Model
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)

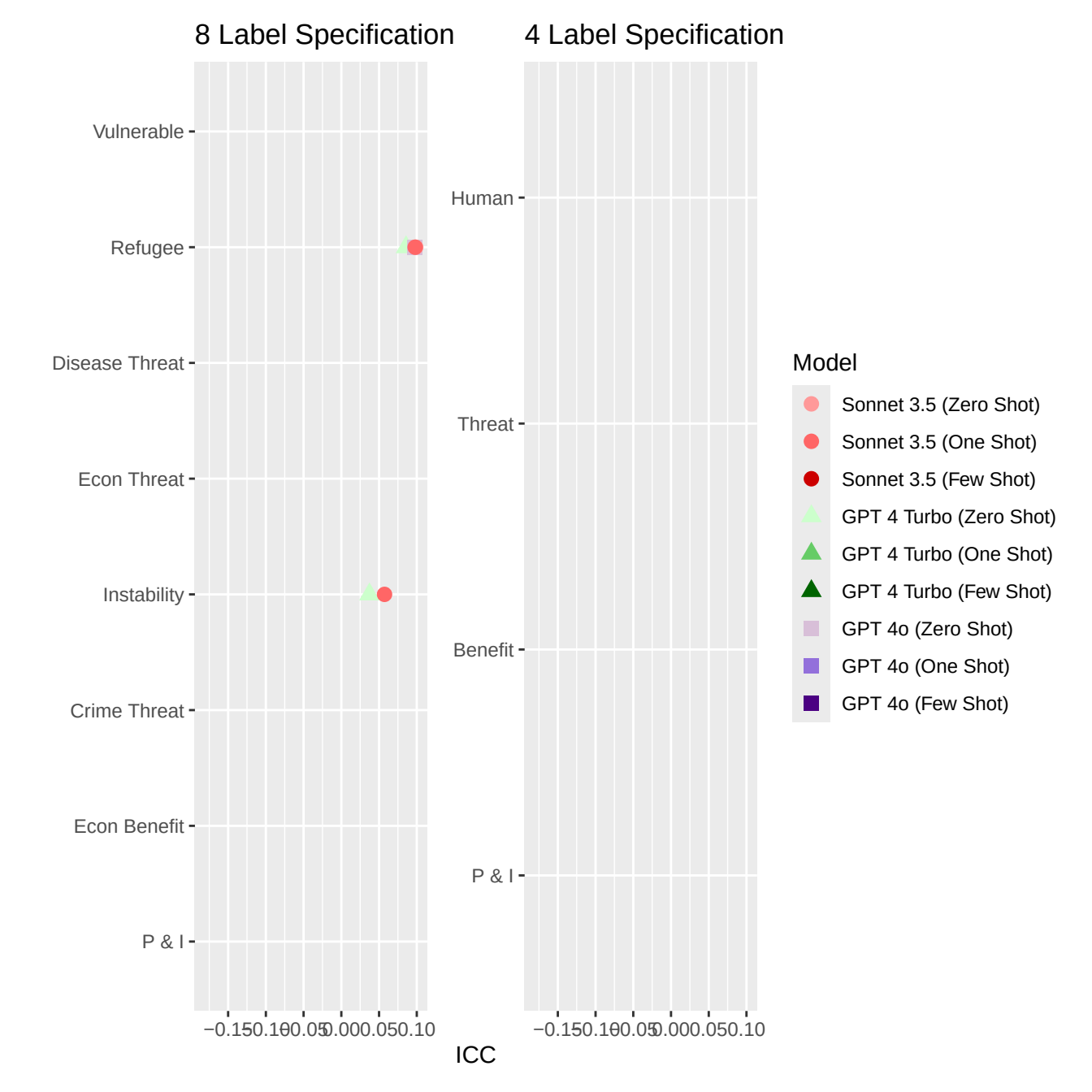

8 Label Specification
4 Label Specification
Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
Human
Threat
Benefit
P & I
0.5
0.6
0.7
0.8
0.9
1.0
Recall
Model
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)

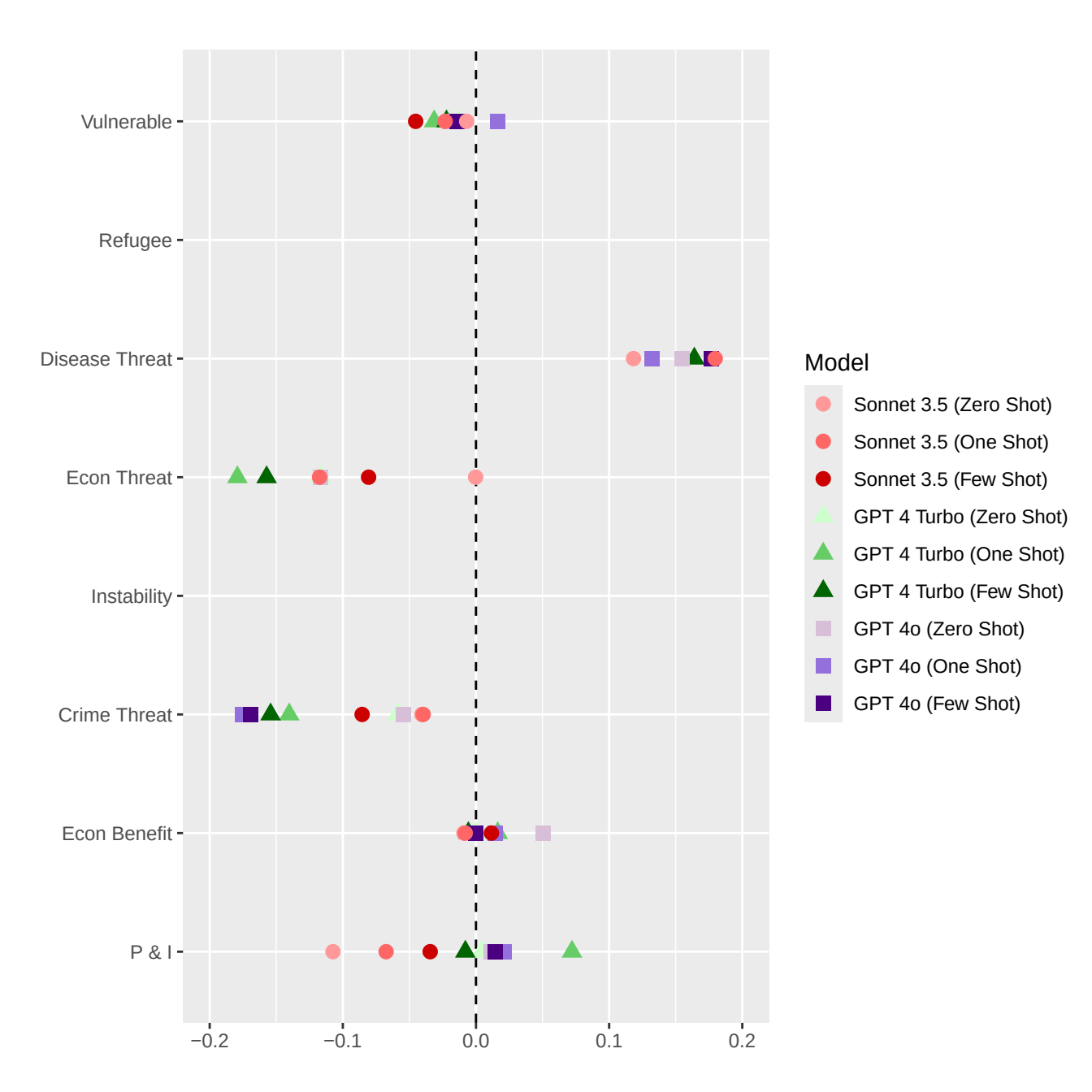

Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
−0.2
−0.1
0.0
0.1
0.2
0.3
Model
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)

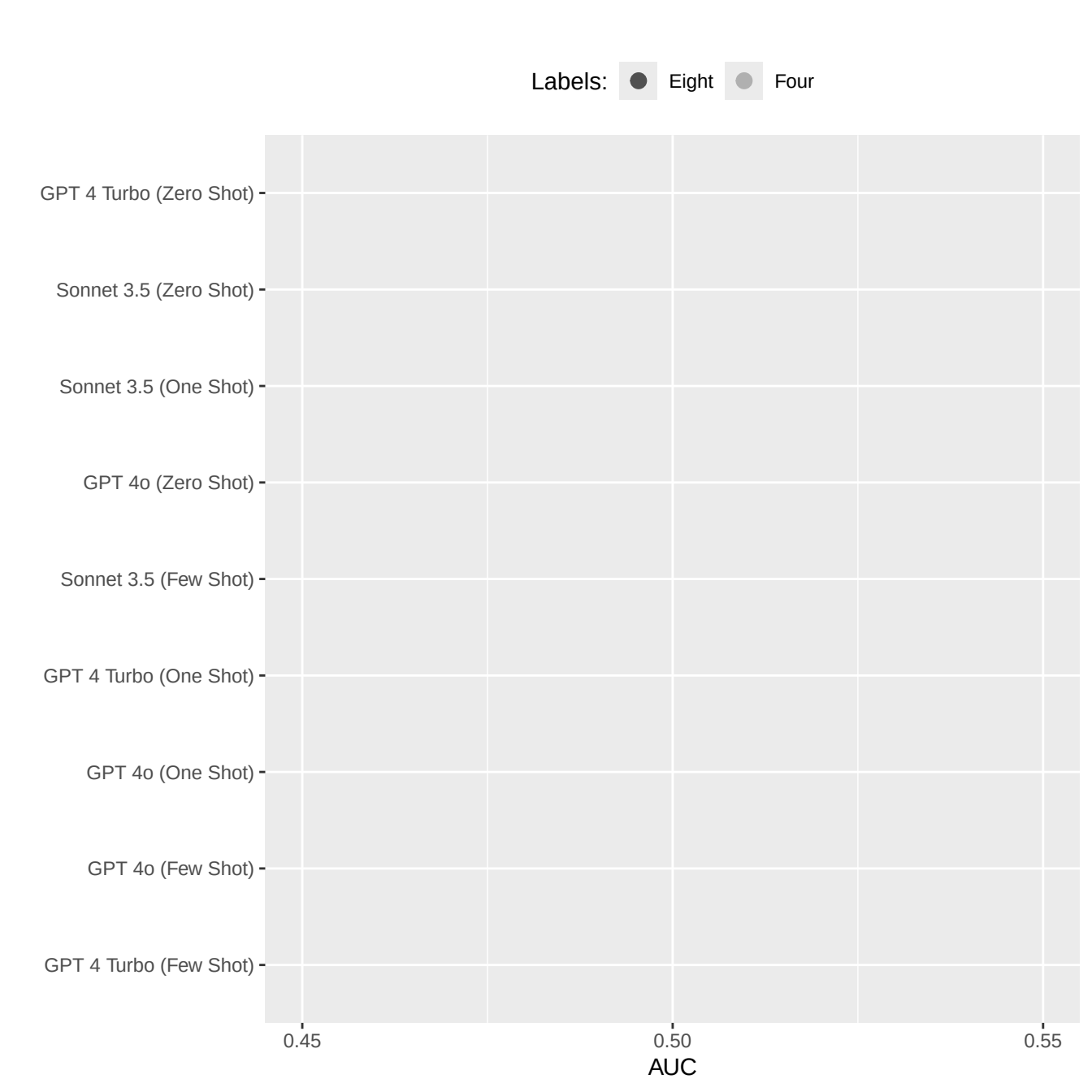

Labels:
Eight
Four
GPT 4o (Few Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (One Shot)
GPT 4 Turbo (One Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Few Shot)
GPT 4o (Zero Shot)
GPT 4 Turbo (Zero Shot)
Sonnet 3.5 (Zero Shot)
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
ICC

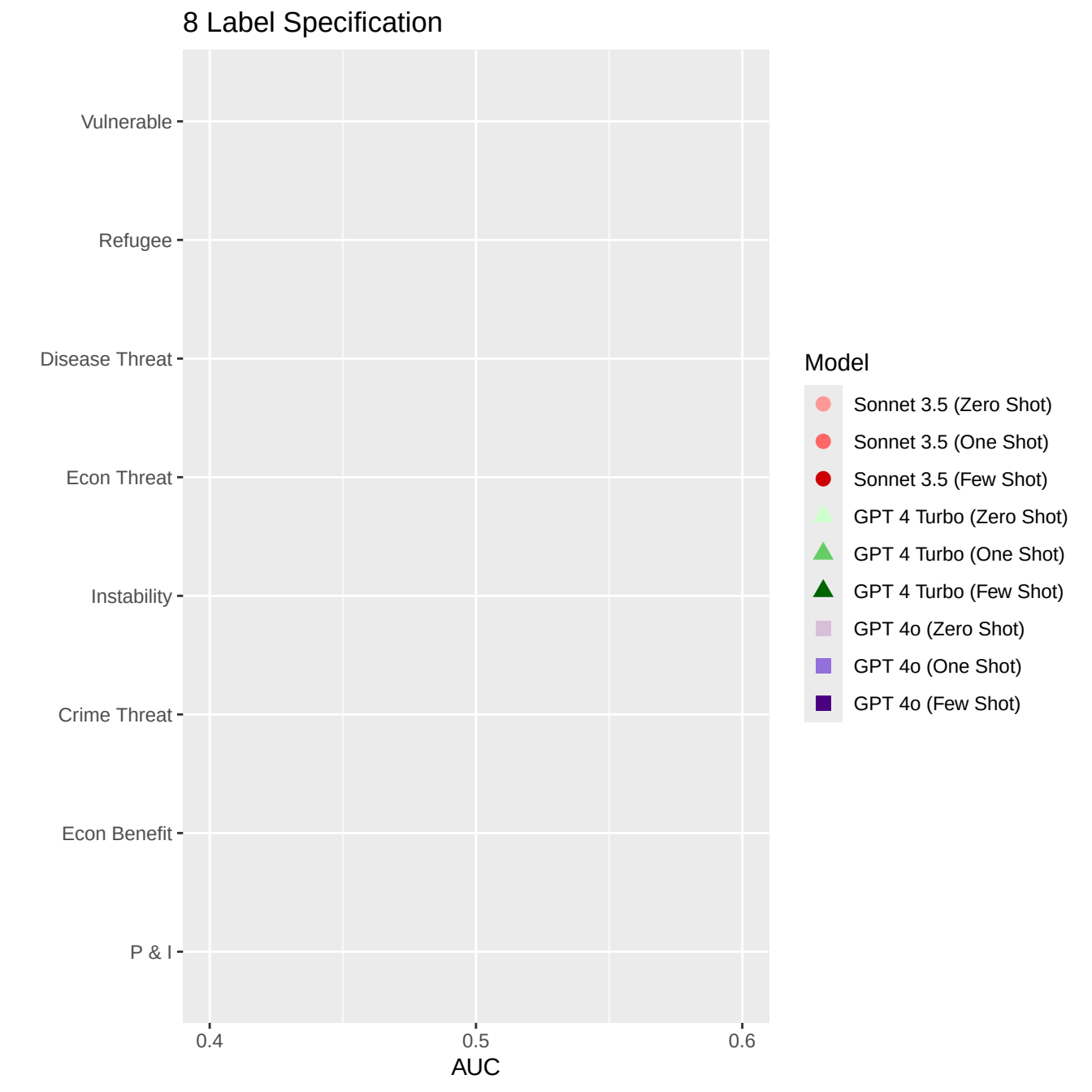

8 Label Specification
Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
−0.15
−0.10
−0.05
0.00
0.05
0.10
ICC
Model
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)

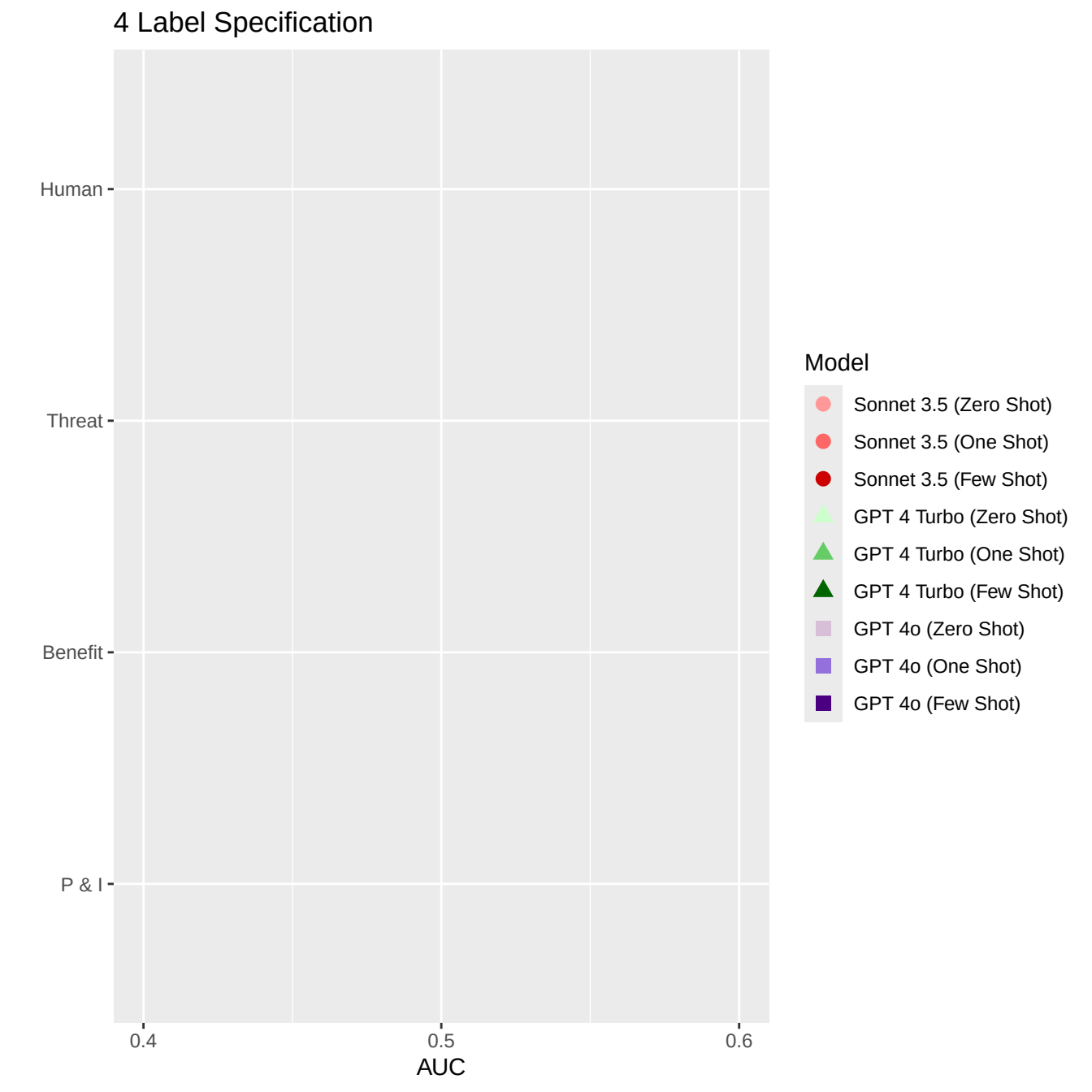

4 Label Specification
Human
Threat
Benefit
P & I
−0.15
−0.10
−0.05
0.00
0.05
0.10
ICC
Model
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)

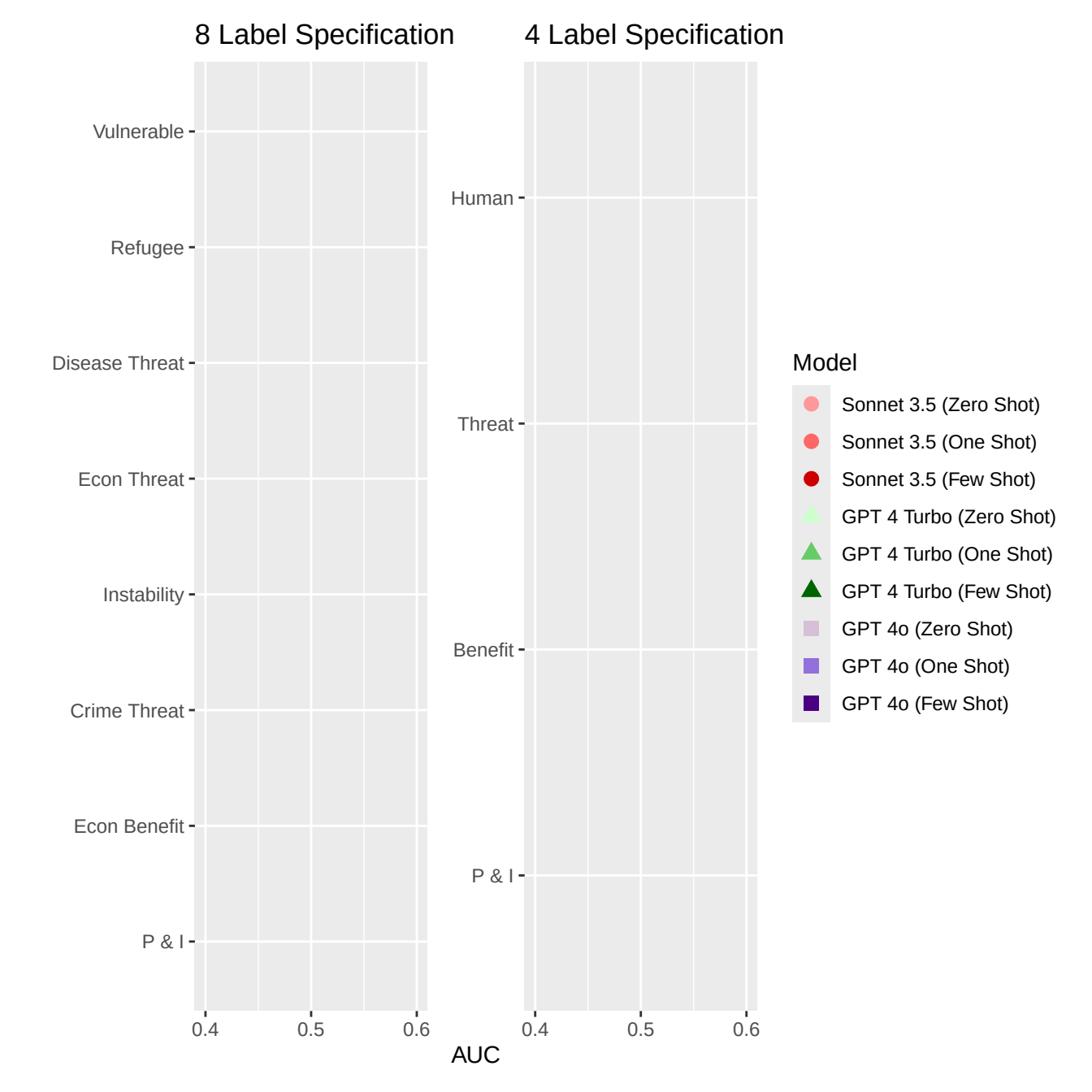

8 Label Specification
4 Label Specification
Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
Human
Threat
Benefit
P & I
−0.15
−0.10
−0.05
0.00
0.05
0.10
ICC
Model
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)

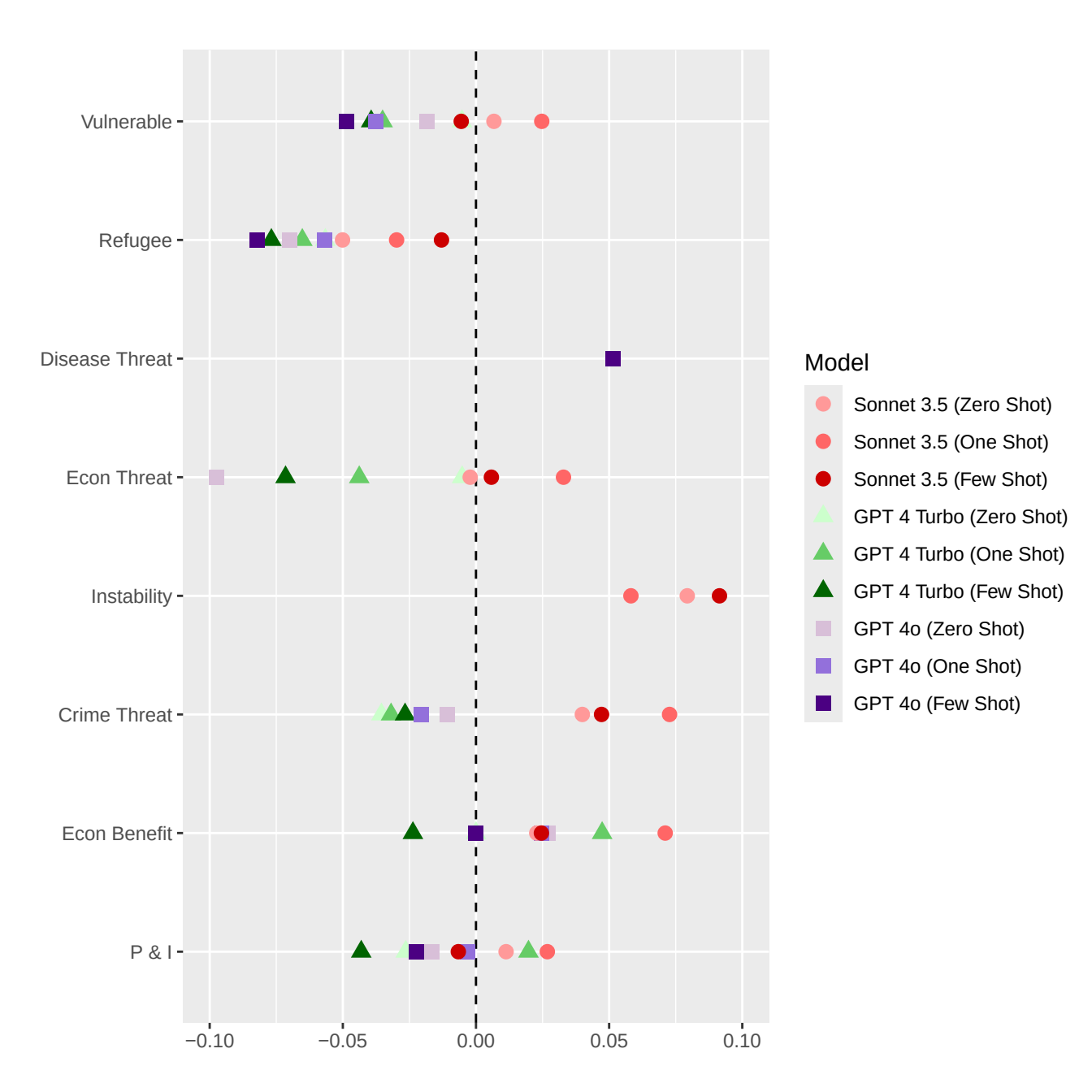

Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
−0.2
−0.1
0.0
0.1
0.2
Model
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)

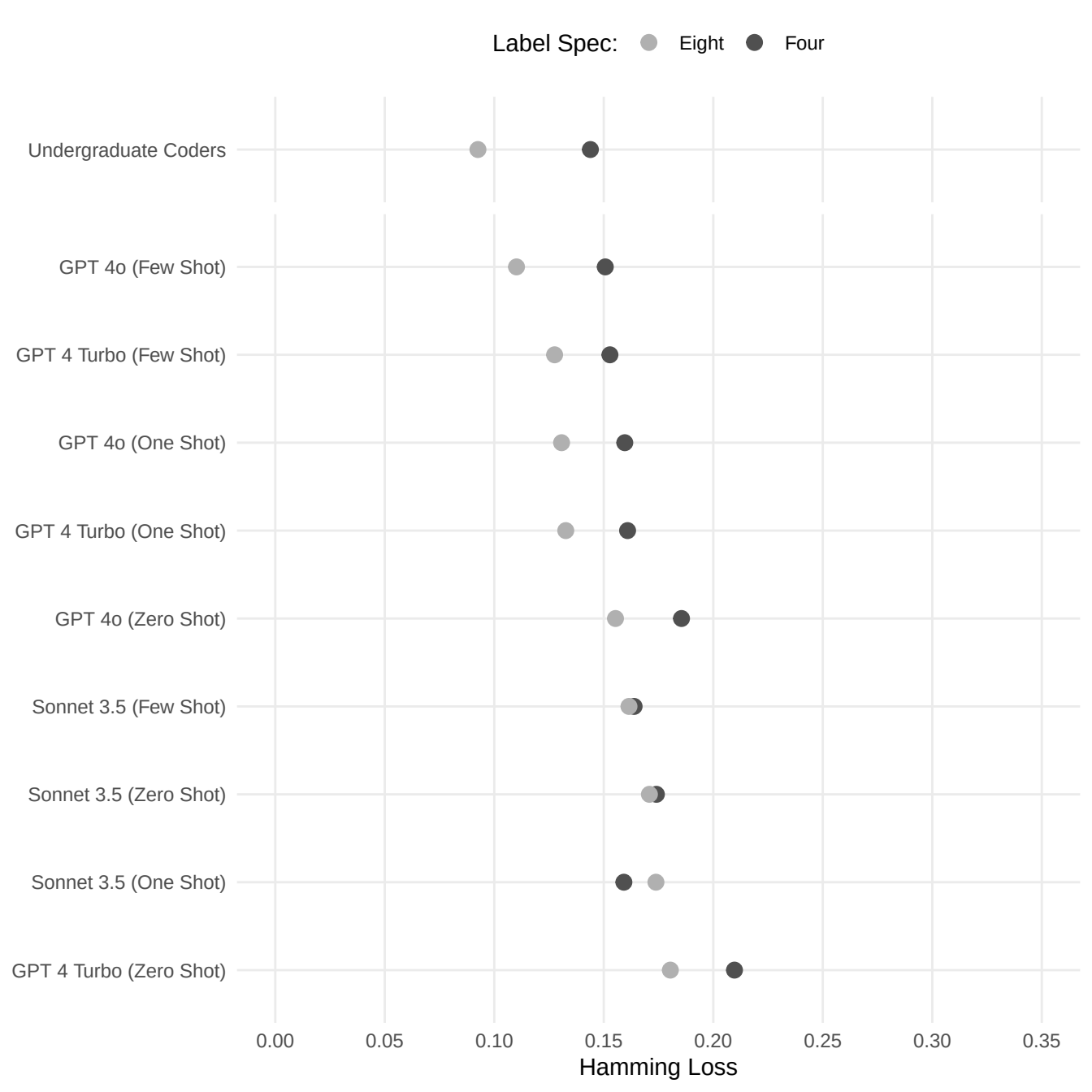

Labels:
Eight
Four
GPT 4 Turbo (Zero Shot)
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
GPT 4o (Zero Shot)
Sonnet 3.5 (Few Shot)
GPT 4 Turbo (One Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)
GPT 4 Turbo (Few Shot)
0.45
0.50
0.55
AUC

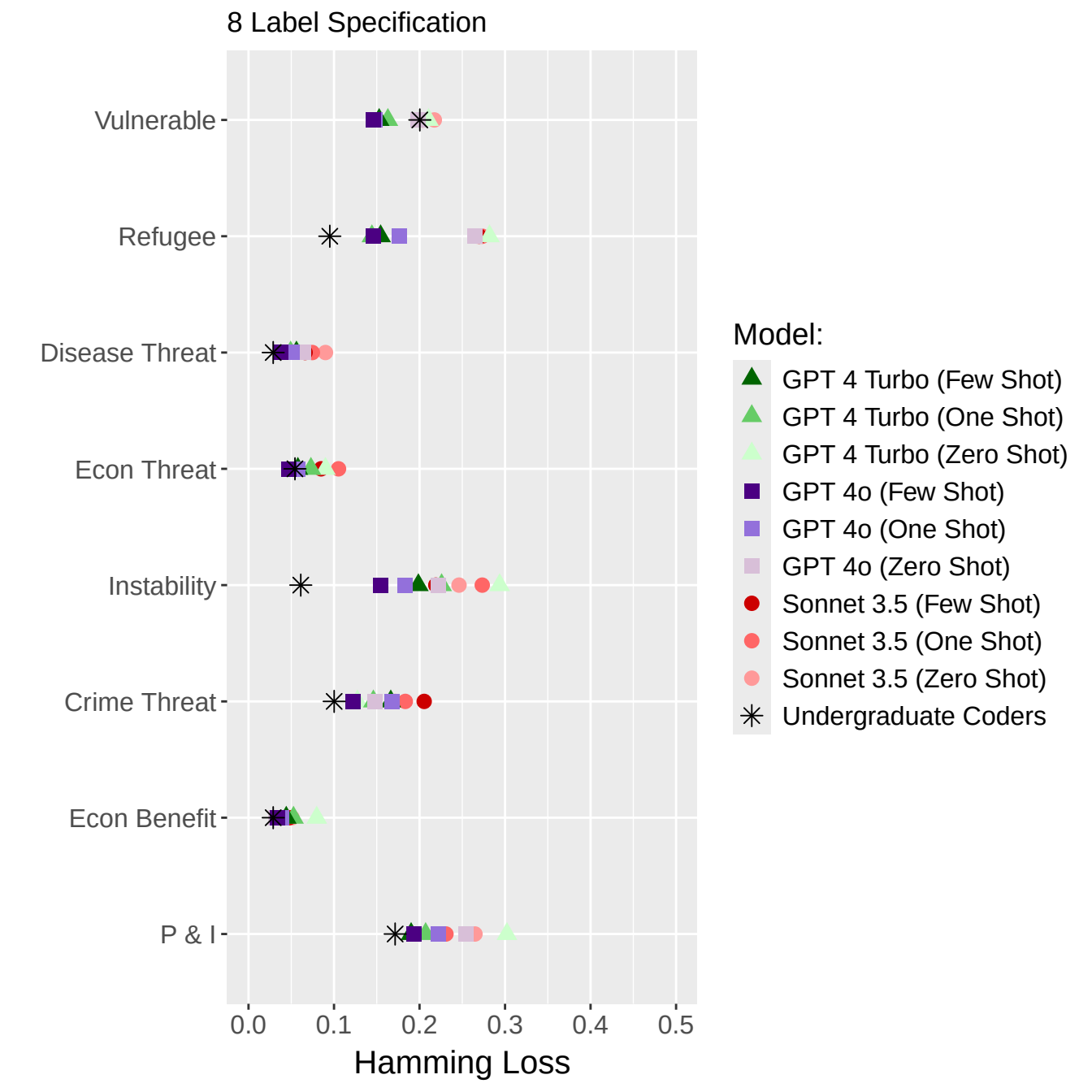

8 Label Specification
Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
0.4
0.5
0.6
AUC
Model
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)

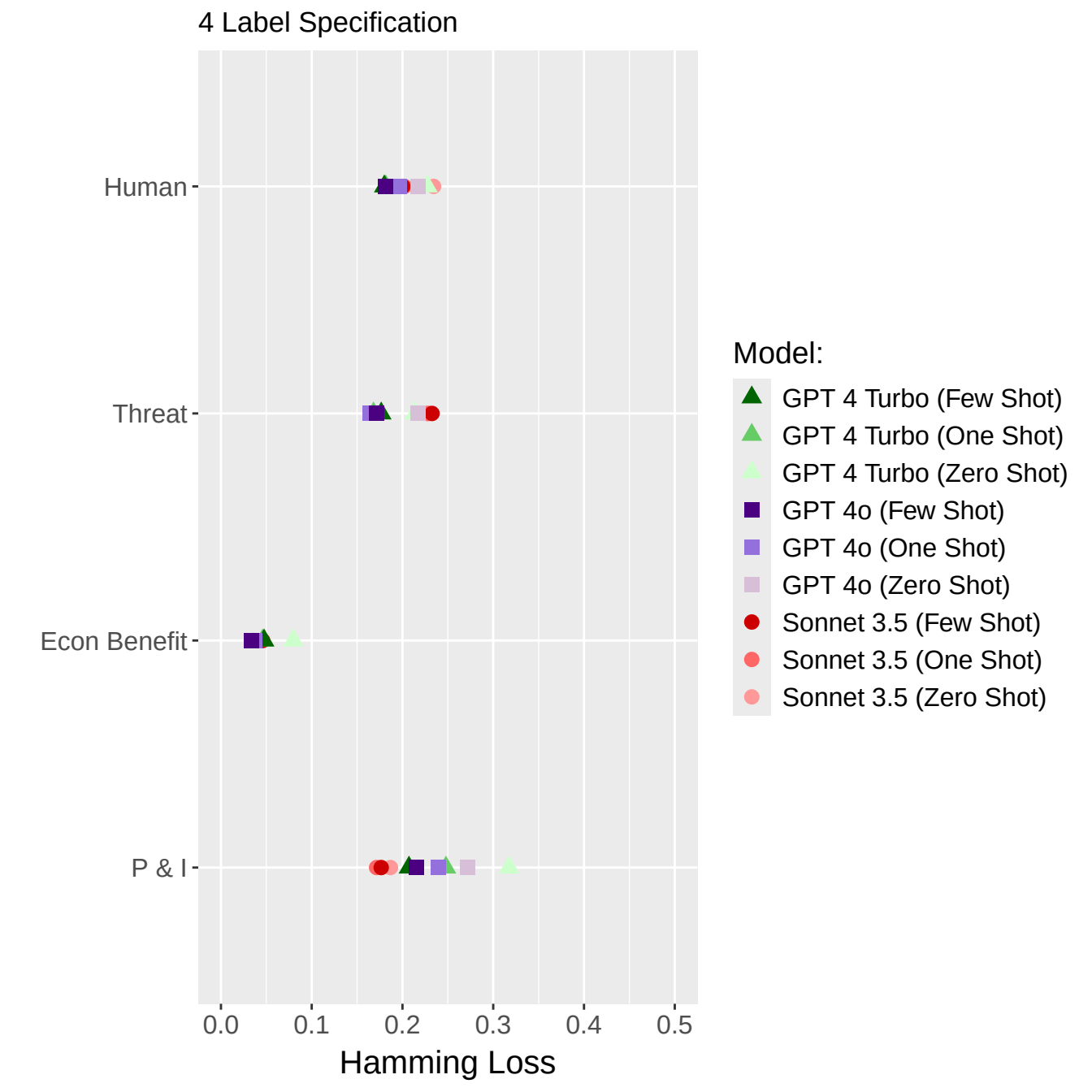

4 Label Specification
Human
Threat
Benefit
P & I
0.4
0.5
0.6
AUC
Model
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)

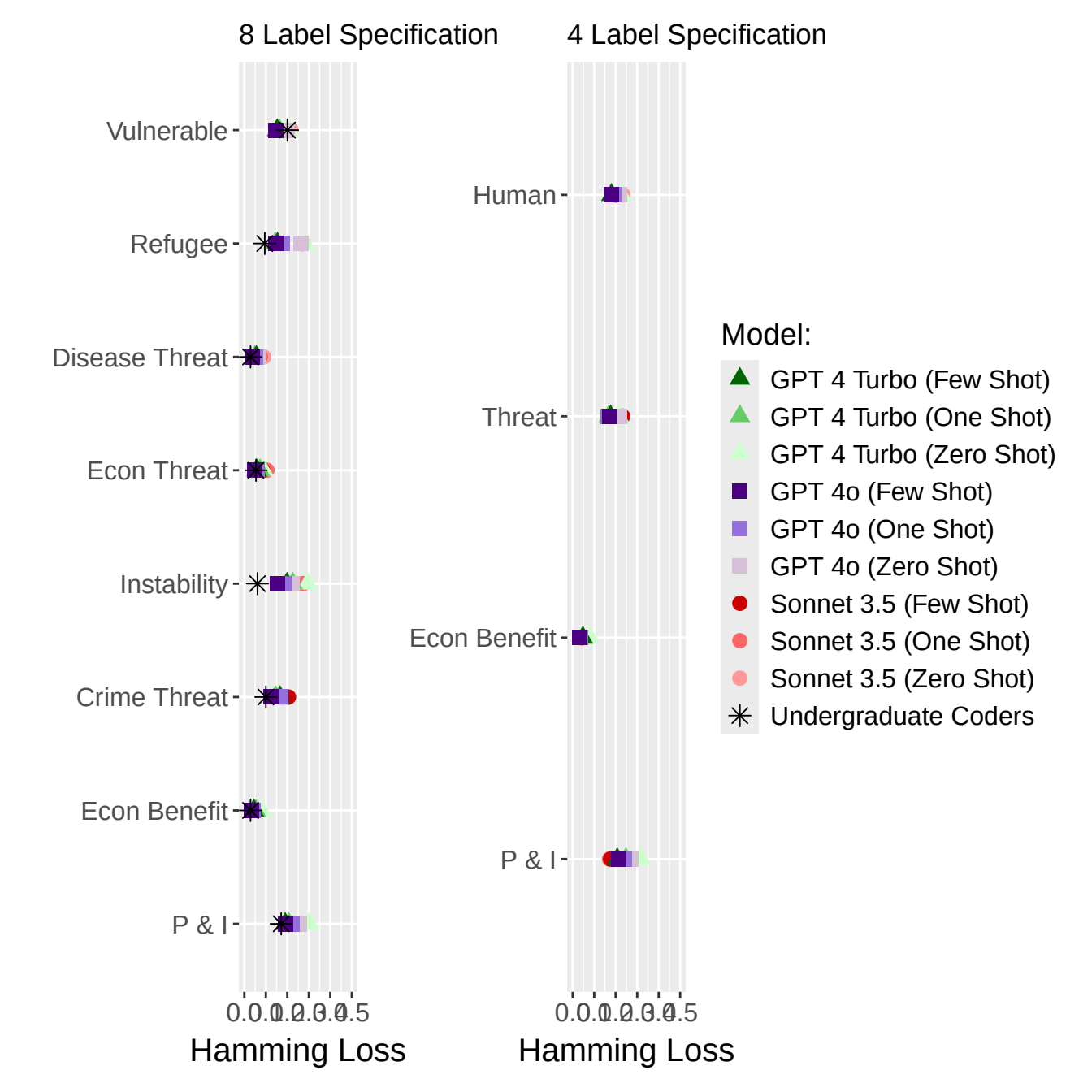

8 Label Specification
4 Label Specification
Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
Human
Threat
Benefit
P & I
0.4
0.5
0.6
AUC
Model
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)

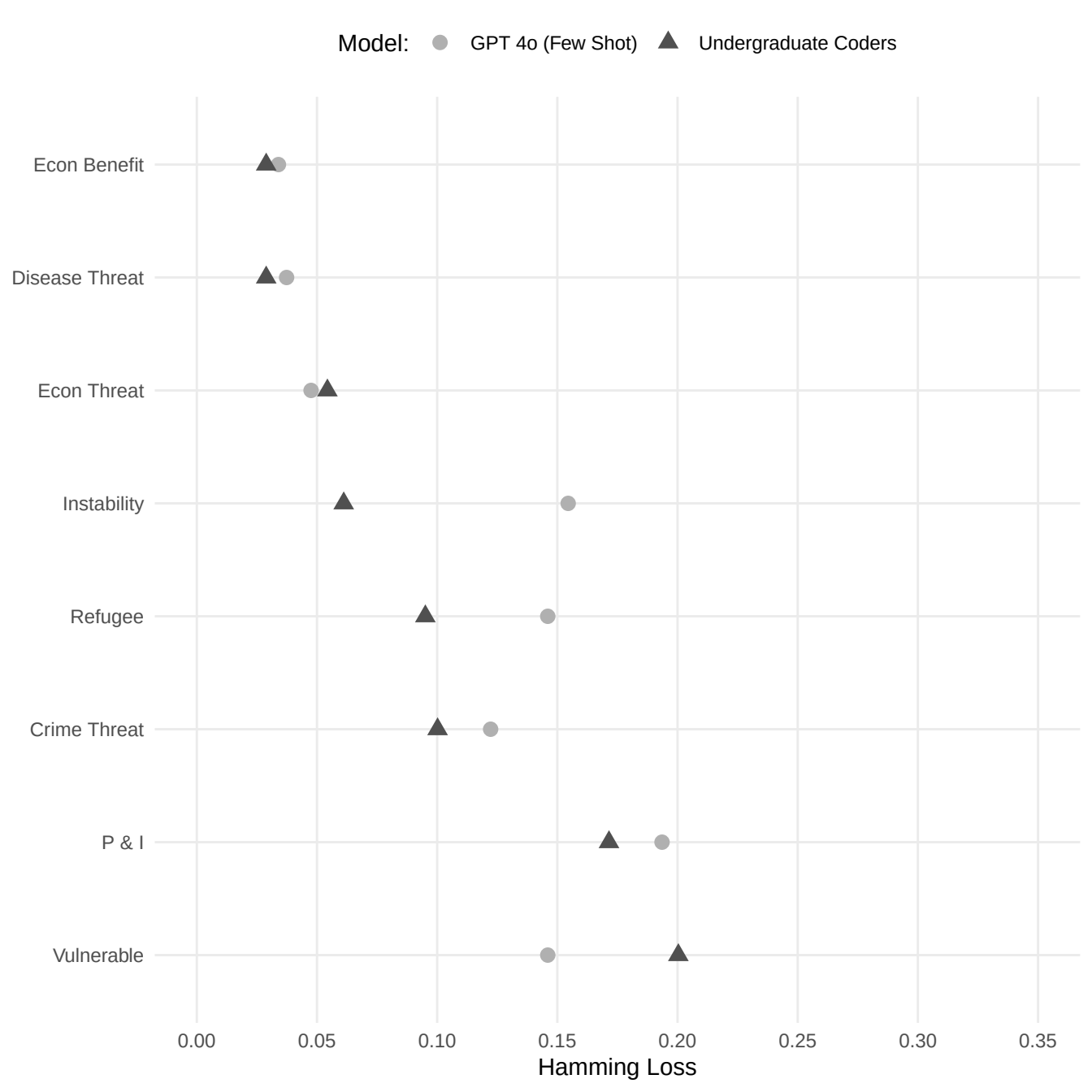

Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
−0.10
−0.05
0.00
0.05
0.10
Model
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Few Shot)
GPT 4 Turbo (Zero Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (Zero Shot)
GPT 4o (One Shot)
GPT 4o (Few Shot)

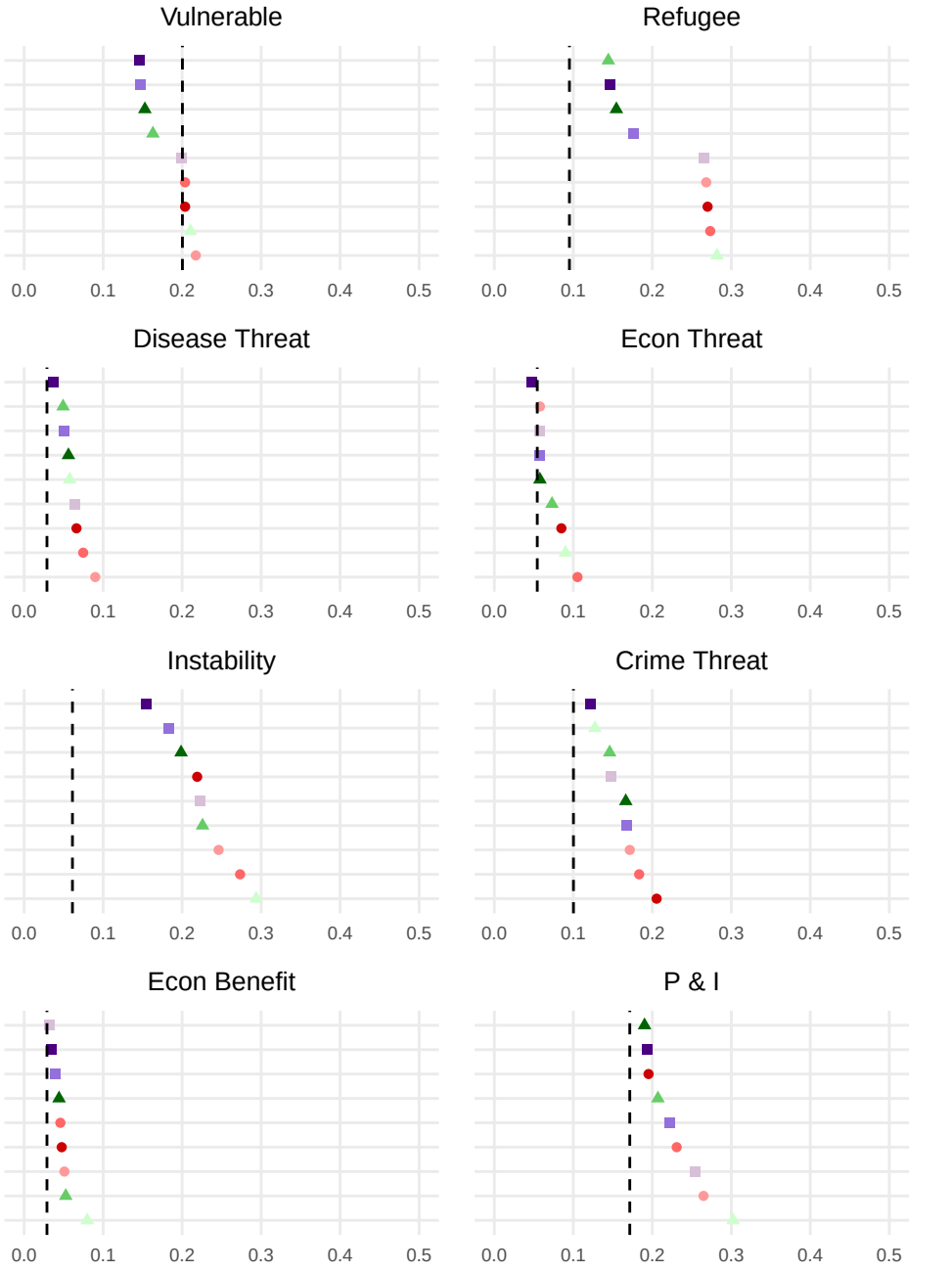

Label Spec:
Eight
Four
Undergraduate Coders
GPT 4o (Few Shot)
GPT 4 Turbo (Few Shot)
GPT 4o (One Shot)
GPT 4 Turbo (One Shot)
GPT 4o (Zero Shot)
Sonnet 3.5 (Few Shot)
Sonnet 3.5 (Zero Shot)
Sonnet 3.5 (One Shot)
GPT 4 Turbo (Zero Shot)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Hamming Loss

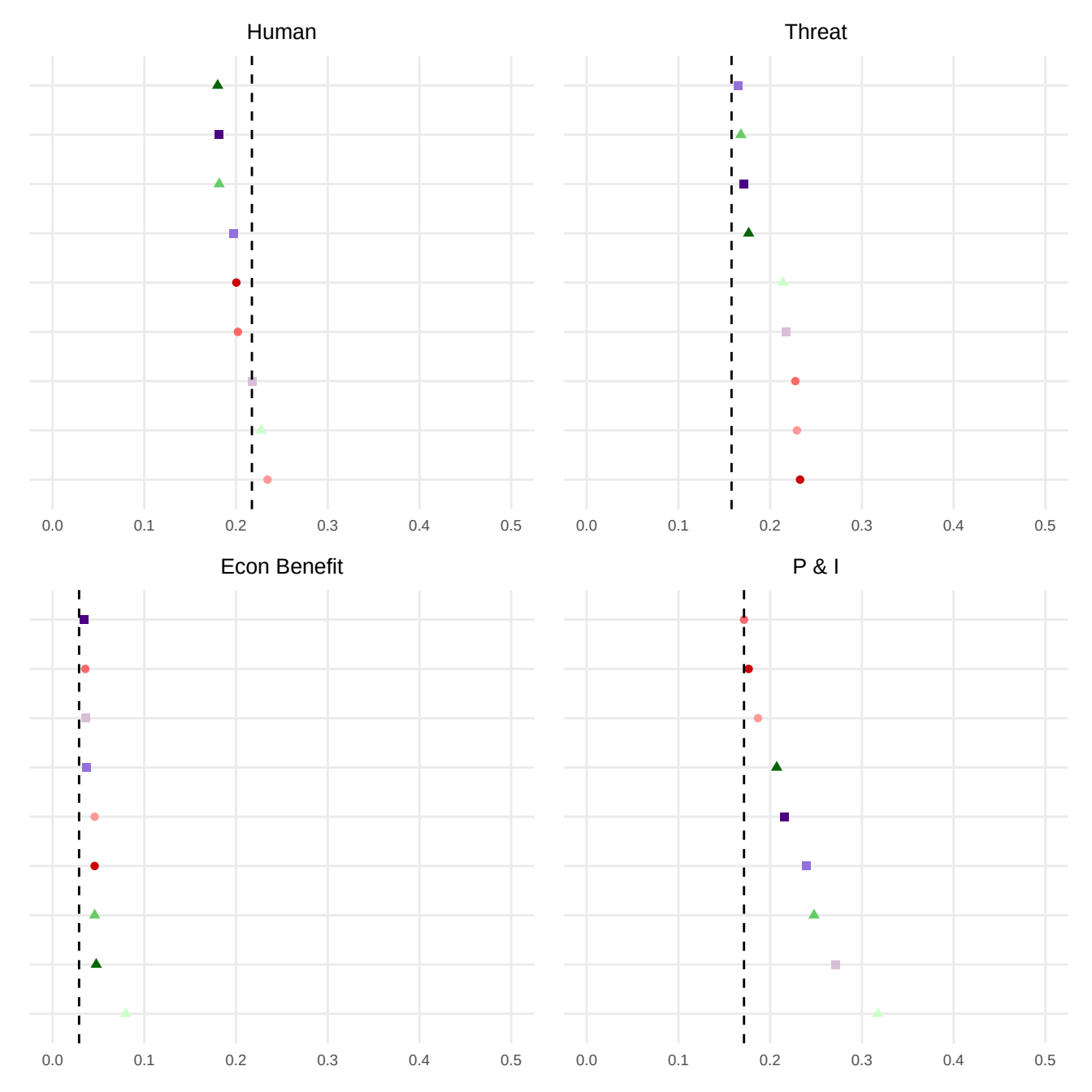

8 Label Specification
Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
0.0
0.1
0.2
0.3
0.4
0.5
Hamming Loss
Model:
GPT 4 Turbo (Few Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Zero Shot)
GPT 4o (Few Shot)
GPT 4o (One Shot)
GPT 4o (Zero Shot)
Sonnet 3.5 (Few Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Zero Shot)
Undergraduate Coders

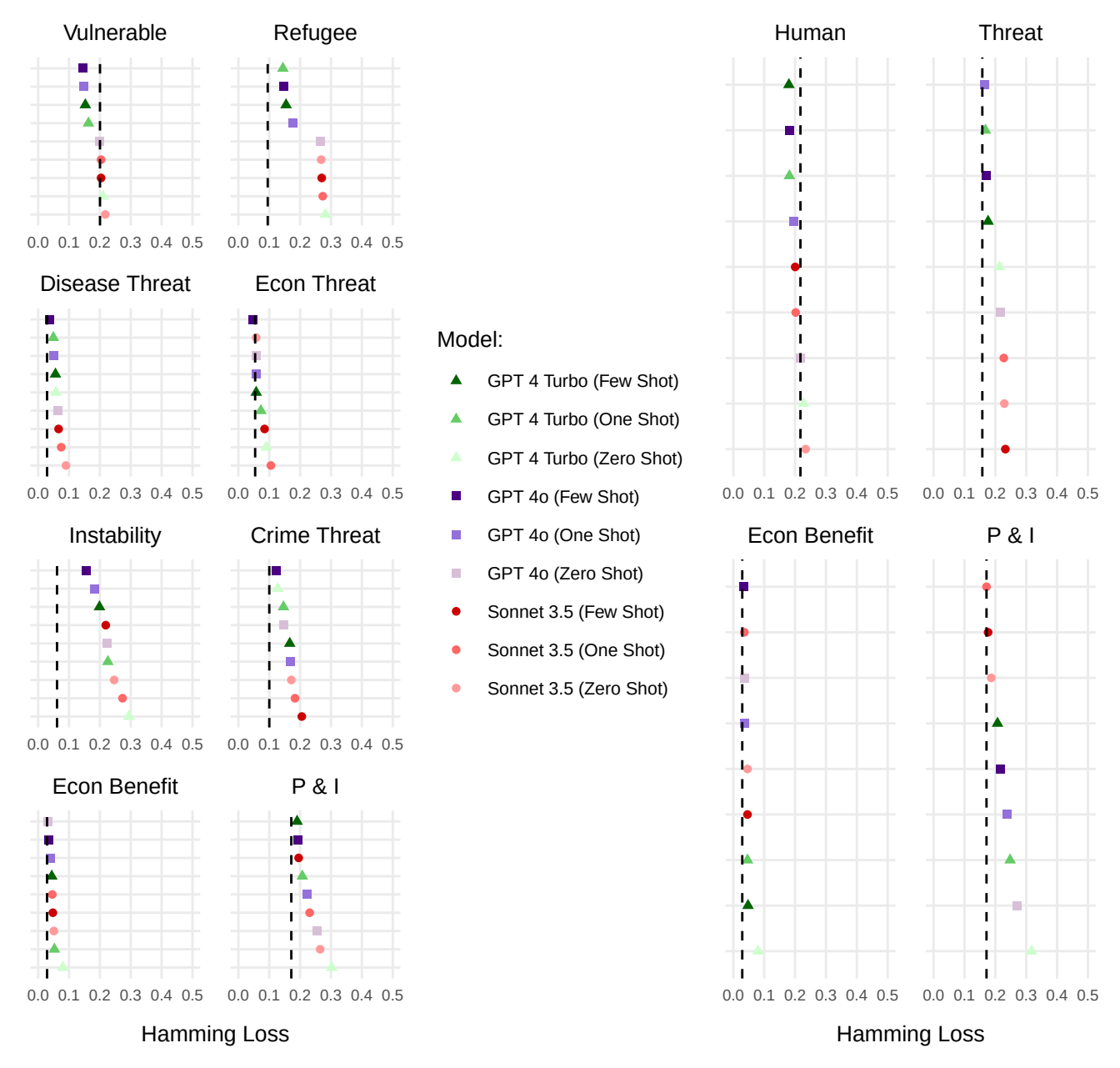

4 Label Specification
Human
Threat
Econ Benefit
P & I
0.0
0.1
0.2
0.3
0.4
0.5
Hamming Loss
Model:
GPT 4 Turbo (Few Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Zero Shot)
GPT 4o (Few Shot)
GPT 4o (One Shot)
GPT 4o (Zero Shot)
Sonnet 3.5 (Few Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Zero Shot)

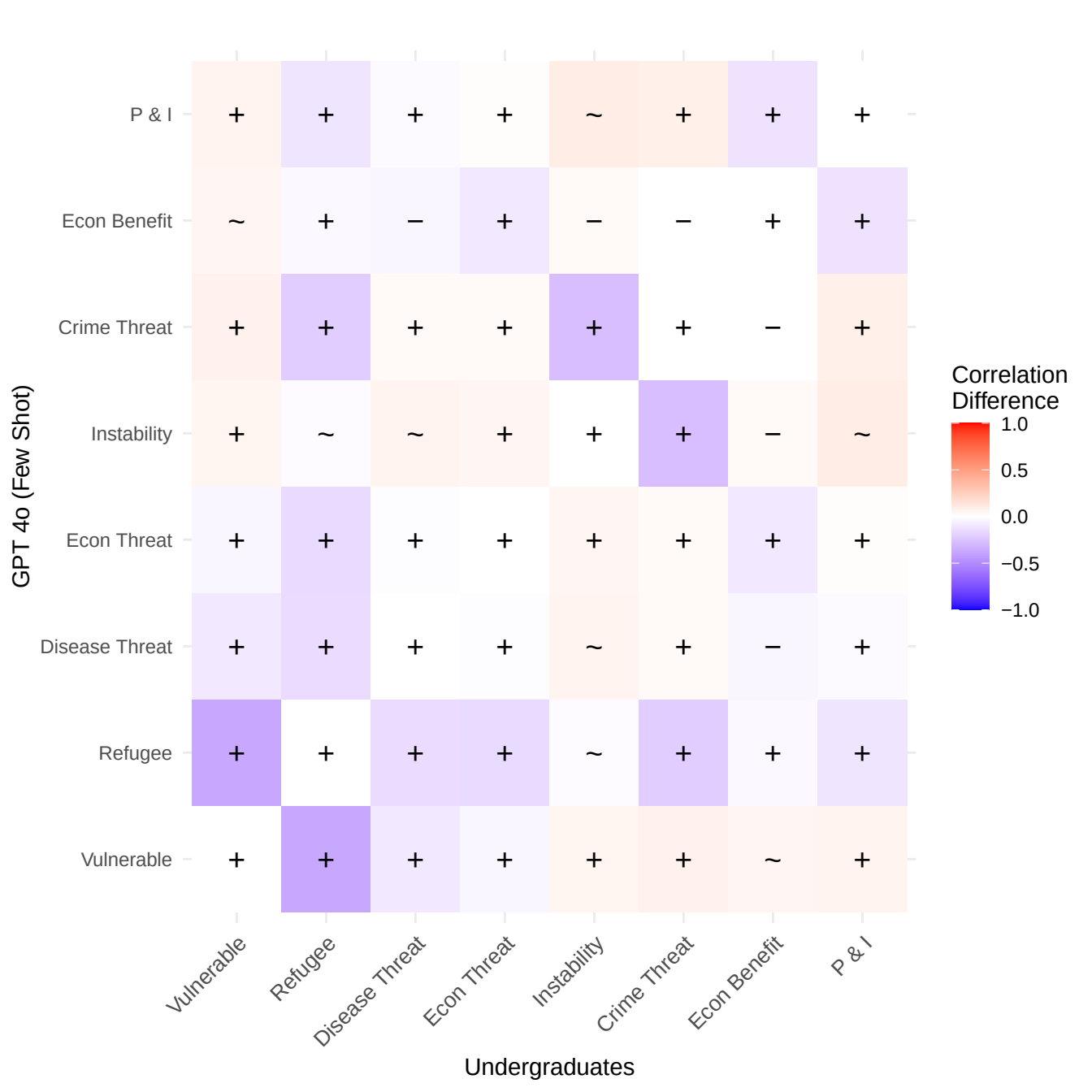

8 Label Specification
4 Label Specification
Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
Human
Threat
Econ Benefit
P & I
0.0
0.1
0.2
0.3
0.4
0.5
Hamming Loss
Hamming Loss
Model:
GPT 4 Turbo (Few Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Zero Shot)
GPT 4o (Few Shot)
GPT 4o (One Shot)
GPT 4o (Zero Shot)
Sonnet 3.5 (Few Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Zero Shot)
Undergraduate Coders

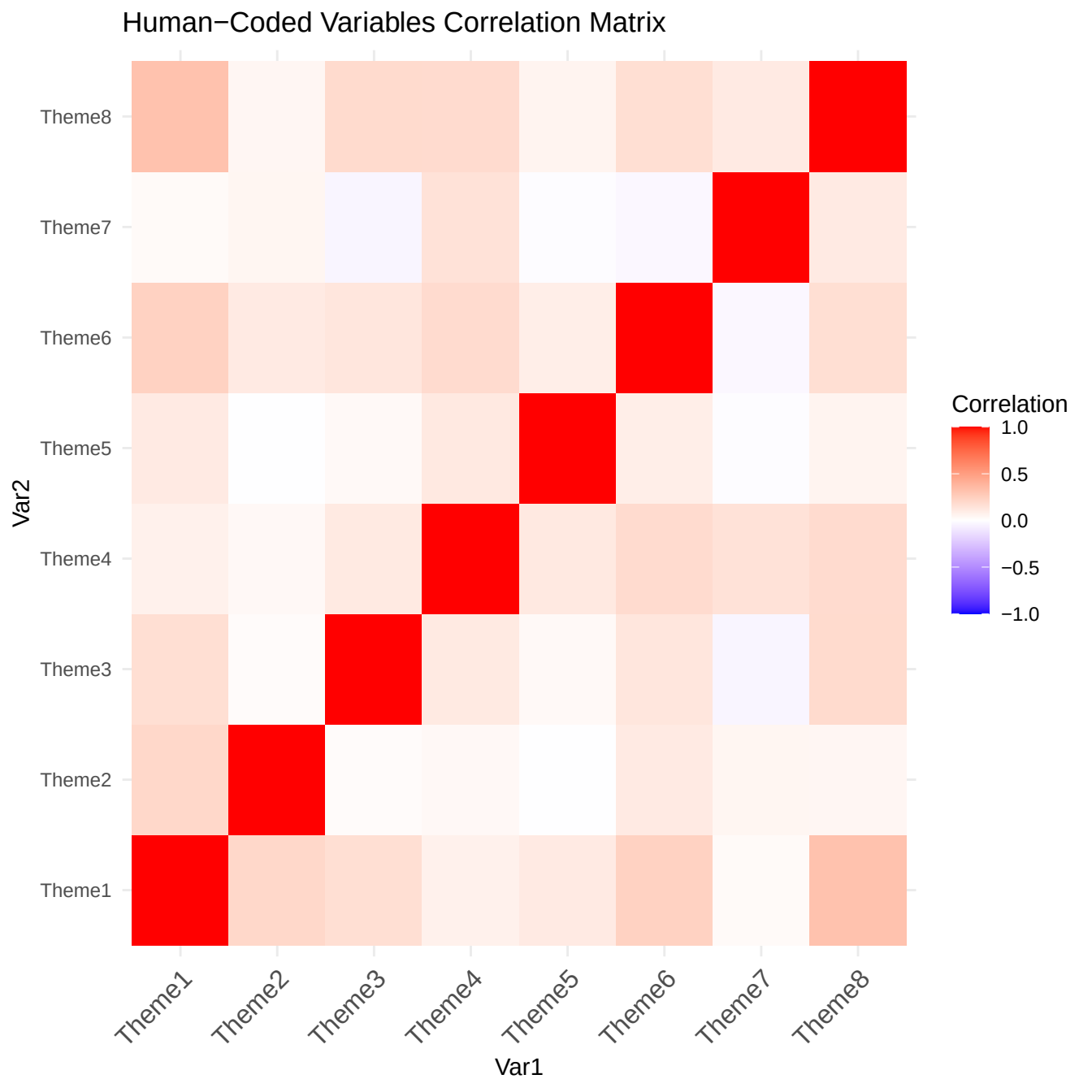

Model:
GPT 4o (Few Shot)
Undergraduate Coders
Econ Benefit
Disease Threat
Econ Threat
Instability
Refugee
Crime Threat
P & I
Vulnerable
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Hamming Loss

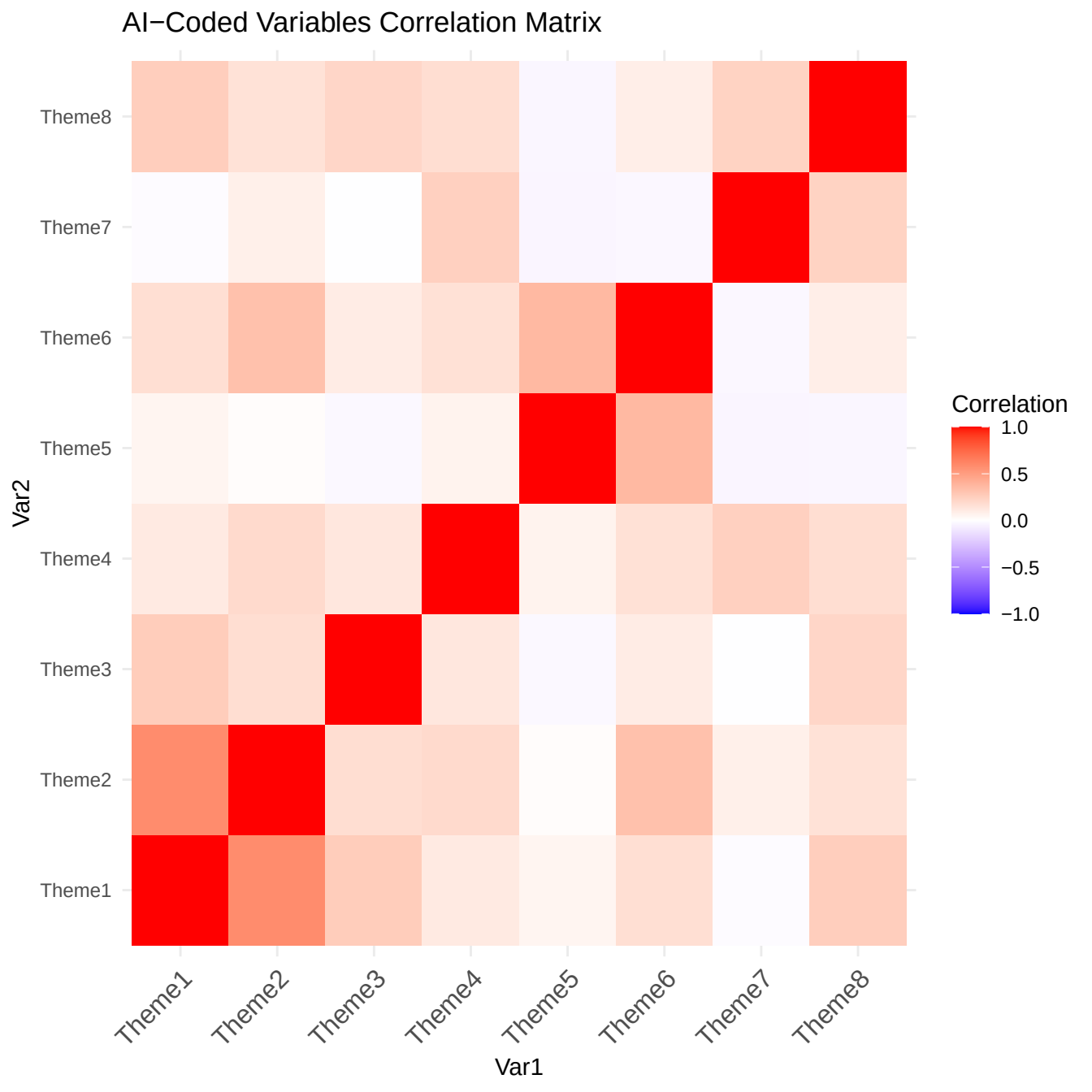

Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
0.0
0.1
0.2
0.3
0.4
0.5
Model:
GPT 4 Turbo (Few Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Zero Shot)
GPT 4o (Few Shot)
GPT 4o (One Shot)
GPT 4o (Zero Shot)
Sonnet 3.5 (Few Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Zero Shot)


Human
Threat
Econ Benefit
P & I
0.0
0.1
0.2
0.3
0.4
0.5


Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
Human
Threat
Econ Benefit
P & I
0.0 0.1 0.2 0.3 0.4 0.5
Model:
GPT 4 Turbo (Few Shot)
GPT 4 Turbo (One Shot)
GPT 4 Turbo (Zero Shot)
GPT 4o (Few Shot)
GPT 4o (One Shot)
GPT 4o (Zero Shot)
Sonnet 3.5 (Few Shot)
Sonnet 3.5 (One Shot)
Sonnet 3.5 (Zero Shot)
Hamming Loss
Hamming Loss


GPT 4o (Few Shot)
P & I
Econ Benefit
Crime Threat
Instability
Econ Threat
Disease Threat
Refugee
Vulnerable
Undergraduates
Vulnerable
Refugee
Disease Threat
Econ Threat
Instability
Crime Threat
Econ Benefit
P & I
Correlation Difference
1.0
0.5
0.0
−0.5
−1.0


Human−Coded Variables Correlation Matrix
Theme8
Theme7
Theme6
Theme5
Theme4
Theme3
Theme2
Theme1
Var2
Theme1
Theme2
Theme3
Theme4
Theme5
Theme6
Theme7
Theme8
Var1
Correlation
1.0
0.5
0.0
−0.5
−1.0


AI−Coded Variables Correlation Matrix
Theme8
Theme7
Theme6
Theme5
Theme4
Theme3
Theme2
Theme1
Var2
Theme1
Theme2
Theme3
Theme4
Theme5
Theme6
Theme7
Theme8
Var1
Correlation
1.0
0.5
0.0
−0.5
−1.0